

CORE-ET CPU Subsystem Micro- Architecture

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Document Control

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Revision History

Version	Date	Author	Notes
v0.1	2026-10-06	Ryan Naveed , Dr. Kamran Malik	• First draft of the document. • Chapter 1, ET-SoC-1 Overview: written, except the Memory Map, which contains a heading only. • Chapter 2, ET-Minion Core: Core Overview (block partitioning, hardware threads, instruction flow) and Front End written. Integer Pipeline, Data Cache and VPU contain headings only. • Chapter 3, Neighborhood: the Overview and the Front End — I-Cache Interface are written. The remaining sections contain headings only.

Source Material and Acknowledgements

The original micro-architecture documents are in [docs/uarch/](#) of this repository, one per module. Where a document and the RTL differ, the RTL is followed.

The engineers who worked on each module are listed below.

Engineer	Module documents
Albert Navarro	FE/Intpipe Description
Francisco Torres	Minion DCache Description
Guillem Solé	Cooperative TensorLoad Description
Ildefonso Gomariz	CORE-ET Neighborhood MAS; Neighborhood I-Cache Description; Front End — I-Cache Interface; CORE-ET Minion Shire Description; Cooperative TensorLoad Description; ET-Link Specification
Jordi Muñoz	Minion VPU Specification
Juanjo Galindo	CORE-ET Minion Shire Description; Minion Shire DLL Constraints
Miquel Izquierdo	ET-Link Specification
Sebastià Tortella	Minion Description; FE/Intpipe Description; ET-Link Specification
Sri Samudrala	Minion VPU Specification

Engineer	Module documents
Xavier Revés	CORE-ET Neighborhood MAS; Minion Description; Minion DCache Description; CORE-ET Minion Shire Description; Minion Shire DLL Constraints; ET-Link Specification

1. ET-SoC-1 Overview

1.1. ET-SoC-1 at a Glance

ET-SoC-1 is a many-core RISC-V system-on-chip developed primarily for artificial-intelligence and machine-learning (AI/ML) workloads. It is designed as a flexible, fast and efficient inference engine for ML applications. Its compute resources are arranged in a multi-level hierarchy that delivers high sustained performance on regular, parallelizable workloads, such as those of ML algorithms.

The device integrates 1,093 general-purpose 64-bit RISC-V cores of three types:

- 1,088 ET-Minion cores for ML computation
- 4 ET-Maxion cores for management and general-purpose tasks
- 1 Service Processor for boot and device management

The cores are connected to on-die SRAM, external memory and a set of standard interfaces through a mesh network-on-chip (NoC). [Table 1](#) summarizes the device.

Table 1. ET-SoC-1 key characteristics

Item	Description
ET-Minion cores	1,088 in-order, dual-threaded RV64IMFC cores, each with an 8-lane vector/tensor unit
ET-Maxion cores	4 single-threaded, superscalar, out-of-order 64-bit RISC-V cores
Service Processor	1 single-threaded core derived from the ET-Minion
On-die SRAM	Each Minion Shire's 4 MB Shire Cache can be partitioned by ESRs into L2, a slice of the chip-wide L3, and scratchpad.
Interconnect	Mesh NoC with 44 mesh stops
External memory	LPDDR4X, 16 channels x 16 bits, up to 4,266 MT/s
PCI Express	Gen4, 8 lanes. Endpoint, root complex, or both at once.
Other interfaces	eMMC, USB 2.0 OTG, I2C/SMBus, I3C, SPI, UART, GPIO. Debug access through JTAG and a USB 2.0 device port.

1.2. The ET-Minion Hierarchy

The ET-Minion cores are organized in three levels, as shown in [Figure 1](#):

- **ET-Minion core**
- **Neighborhood**: 8 ET-Minion cores
- **Minion Shire**: 4 Neighborhoods (32 ET-Minion cores)

Each level adds resources that are shared by the units it contains. The device contains 34 Minion Shires, which together form the Minion Shire array ([Section 1.2.4](#)).

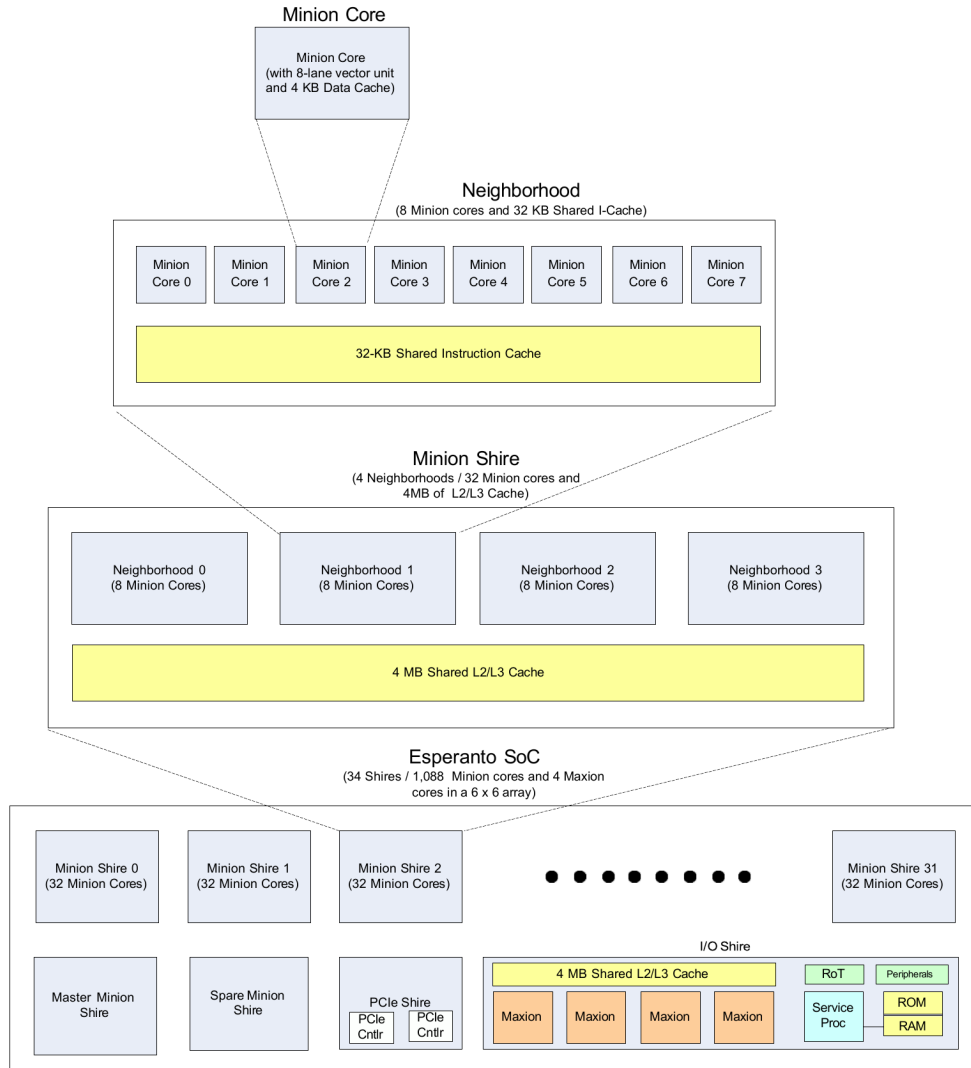


Figure 1. ET-Minion hierarchy

1.2.1. ET-Minion Core

The ET-Minion is a dual-threaded, in-order, single-issue 64-bit RISC-V core that implements RV64IMFC. Its main features are:

- Two hardware threads (harts)
- 4 KB private L1 data cache, partly configurable as scratchpad memory
- Vector processing unit (VPU) with eight identical lanes operating in lock-step. Each lane contains:
 - Floating-point multiply-add unit (FMA): one 32-bit or two 16-bit operations
 - Two integer multiply-add units (IMA): four 8-bit multiply-accumulates each
 - Integer unit (INT)
 - Transcendental unit (TRANS)
- Esperanto instruction extensions for vector, tensor, atomic, message, cache-management and fast-synchronization operations

The core consists of four top-level blocks, shown in [Figure 2](#):

- **Front end (FE)**
- **Integer pipeline**
- **L1 data cache and scratchpad**
- **VPU**

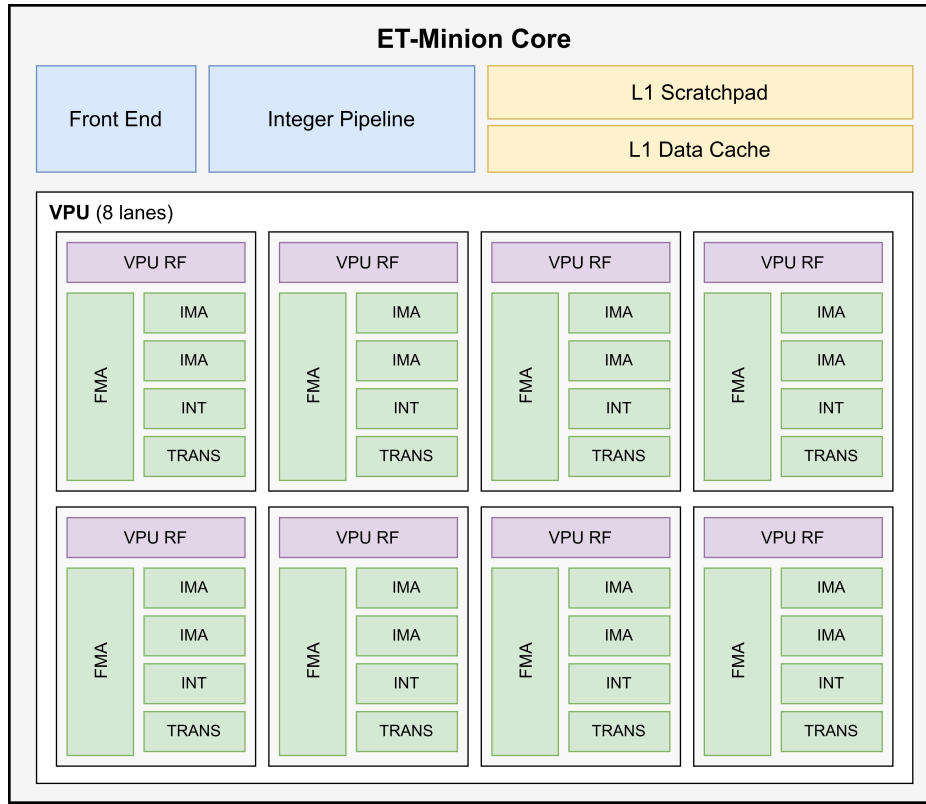


Figure 2. ET-Minion core

1.2.2. Neighborhood

A Neighborhood groups eight ET-Minion cores that share the following resources. In [Figure 3](#), the L0 micro-caches are labeled "I-Cache".

A Neighborhood also spans two supply-voltage regions. The cores and most of the Neighborhood logic are in the low-voltage region, while the interface towards the Shire Channel and the memory cells that need speed, among them the data RAMs of the L1 instruction cache, are in the high-voltage region ([Section 3.1.3](#)).

- **L1 instruction cache:** 32 KB, organized as 128 sets $\times$ 4 ways with 64-byte lines. It is shared by all eight cores, and its data RAMs are located outside the Neighborhood.
- **L0 micro-caches:** two, each shared by four cores. Each holds 16 fully associative cache lines, sits between its cores and the L1 instruction cache, and accesses the L1 instruction cache on a miss.
- **Page-table walkers (PTWs):** two, each shared by four cores. Each PTW serves the data caches of its four cores and the instruction-cache side.
- **Request/response path:** a single path to the memory hierarchy outside the Neighborhood. Requests from the eight data caches, the instruction cache, the two PTWs and, the cooperative TensorLoad/TensorStore units are arbitrated into (256-bit internally which is converted to a 512-bit ET-Link bus).

- **Performance monitoring unit (PMU):** twelve 64-bit event counters shared by the eight cores. Eight counters count ET-Minion events and four count Neighborhood events.
- **Cooperative TensorLoad:** coalesces the TensorLoad requests that several cores issue for the same address into a single memory request, and delivers the one response back to every cooperating core. Cores in all four Neighborhoods of a Minion Shire can cooperate; one Neighborhood acts as master and sends the combined request.
- **Cooperative TensorStore:** combines the partial writes of cooperating cores into one Shire Cache write, and replicates the acknowledgement to every core that took part. Cores cooperate in two fixed groups, cores 0 to 3 and cores 4 to 7, in one of three modes:
 - **Quad-128:** 128 bits from each of four cores, forming a full 512-bit cache line
 - **Pair-256:** 256 bits from each of two cores, forming a full 512-bit cache line
 - **Pair-128:** 128 bits from each of two cores, forming half a cache line

Each core holds its request until every core of the group has presented one, so operations of different modes cannot mix.

- **Fast Local Messaging Network (FLN):** carries messages between cores of the same Neighborhood directly, instead of sending them out through the request path and back in through the response path. Connections are not all-to-all; they follow the pattern used by the tensor reduction instructions, and a message arriving over this network takes priority over the other responses to that core.
- **Fast local barrier (FLB) path**
- **Neighborhood ESRs**
- **Interrupts:** routes the external, timer and software interrupt lines to the cores, and delivers the inter-processor interrupts (IPIs). A software write to a Shire ESR raises a machine software interrupt for a hart, or a redirect IPI that sends the selected harts to the PC held in the Neighborhood `ipi_redirect_pc` ESR.

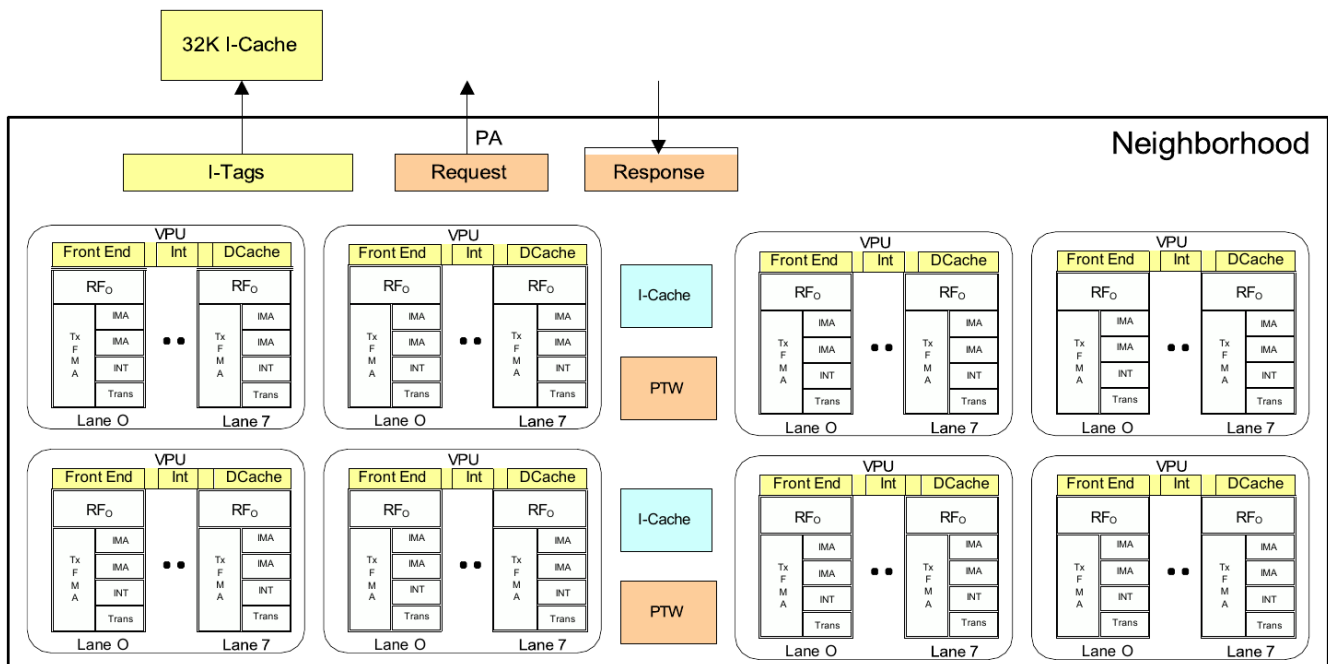


Figure 3. Neighborhood

1.2.3. Minion Shire

A Minion Shire groups four Neighborhoods (Figure 4). It also contains:

- **Shire Cache:** 4 MB, in four 1 MB banks. Programmable ESRs divide the cache into three kinds of partition:
 - L2 cache, private to the Shire
 - Slice of the chip-wide distributed L3 cache
 - Scratchpad memory
- **Crossbar:** Request and response crossbars connect four Neighborhoods and the RBOX port to the four Shire Cache banks and the UC block. Each Neighborhood has one request and response port into the Shire Cache, and each bank accepts at most one request per cycle. The crossbar also connects the Neighborhoods to the un-cached (UC) block.
- **Uncached (UC) block:** receives the Neighborhood requests that are not sent to the Shire Cache banks. It handles uncacheable accesses, contains the fast local barrier (FLB) and fast local credit counter (FCC) units, and delivers messages sent between ET-Minion cores of different Shires.
- **ESR banks:** the Esperanto System Registers (ESRs) configure the blocks in the Shire.
- **NoC interface:** L2 misses and uncacheable accesses leave the Shire here, bound for other Shires or the memory controllers.

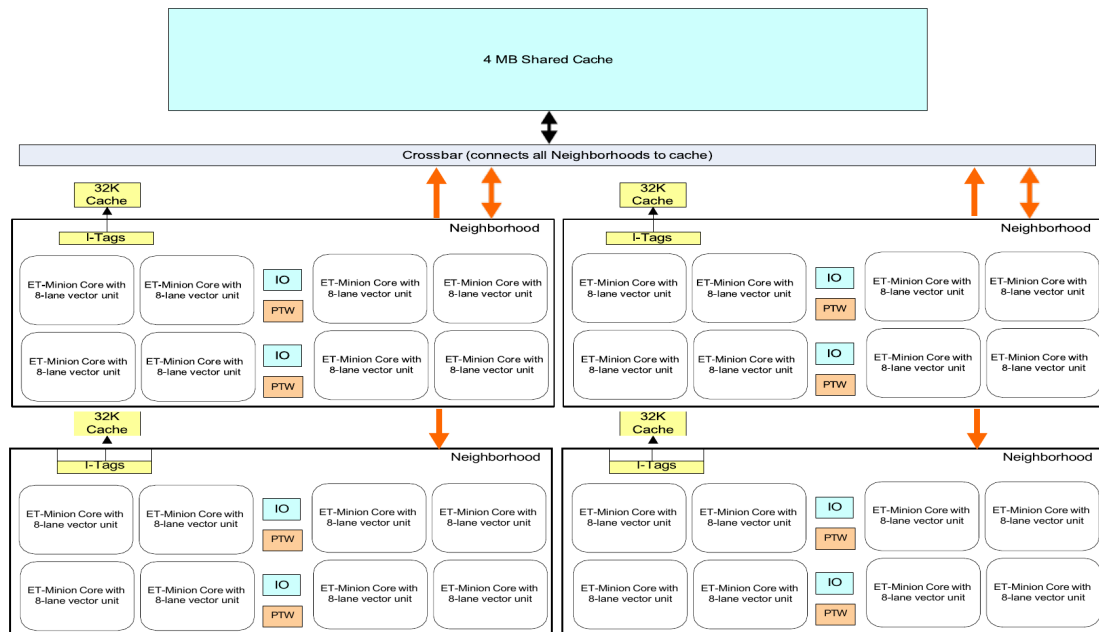


Figure 4. Minion Shire

1.2.4. Minion Shire Array

The 34 Minion Shires, 1,088 ET-Minion cores in total, form the Minion Shire array. The roles of the Shires are assigned by software:

- **Compute:** 32 Minion Shires operate in parallel as the compute array.
- **Master (management):** one Minion Shire receives work and distributes it to the compute array.

- **Spare:** one Minion Shire.

All Minion Shires are identical in hardware, so any Minion Shire can take any of these roles.

1.3. ET-Maxion and Service Processor

In addition to the ET-Minion cores, the device contains two other core types. Both are located in the I/O Shire ([Section 1.4.4](#)). This document does not describe them beyond this section.

1.3.1. ET-Maxion Cores

The ET-Maxion is a single-threaded, superscalar, out-of-order 64-bit RISC-V core intended for management and general-purpose tasks. The device contains four identical ET-Maxion cores. Together with a 4 MB shared L2/L3 cache they form the Maxion Neighborhood, and a system bus and coherency hub in the I/O Shire connects them. [Figure 5](#) shows the ET-Maxion pipeline.

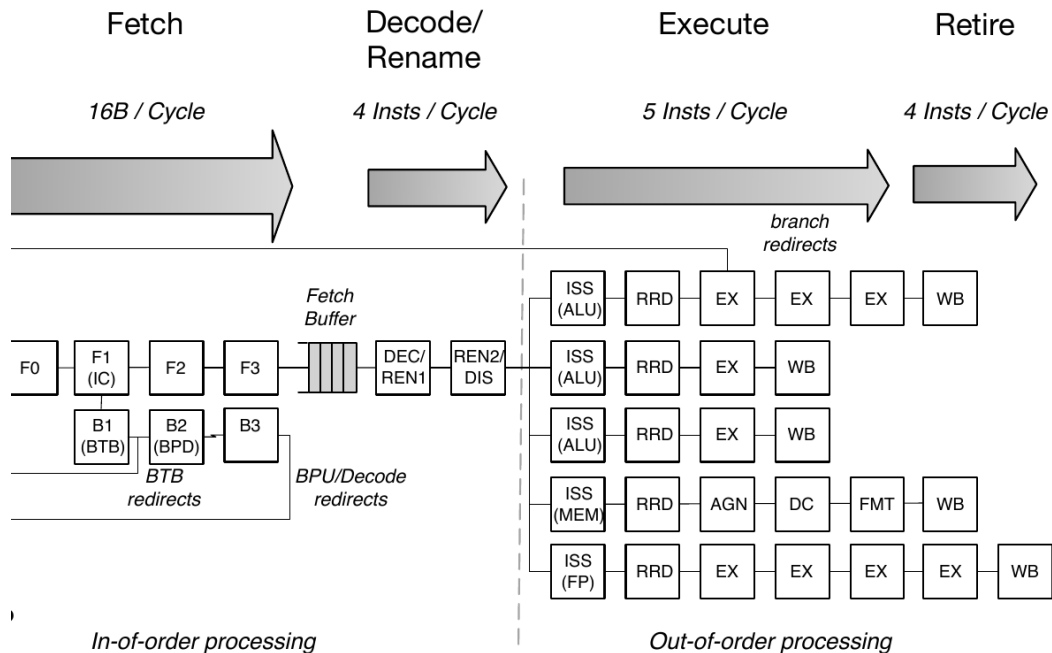


Figure 5. ET-Maxion pipeline

1.3.2. Service Processor

The Service Processor (SP) is a single-threaded ET-Minion core without the vector/tensor extensions. It has its own ROM and a 1 MB scratchpad SRAM, but none of the Neighborhood or Shire shared resources.

- **Boot.** The SP is the first major block released from reset. It executes the boot code from ROM and initializes the device interfaces, including DDR, PCIe and USB.
- **Run time.** It provides management services for the whole device.
- **Access.** The SP can access every resource on the device, and some resources are accessible only to it.

The I/O Shire also contains a hardware root of trust, and six secure mailboxes for communication

between four agents:

- Service Processor
- Master Minion Shire
- ET-Maxions
- PCIe host

1.4. Mesh Network-on-Chip

The mesh NoC is the top-level interconnect of ET-SoC-1 and connects all Shires. [Figure 6](#) shows the device with the Shire array on the mesh, the external memory and PCIe interfaces, and the peripheral, Service Processor and debug networks.

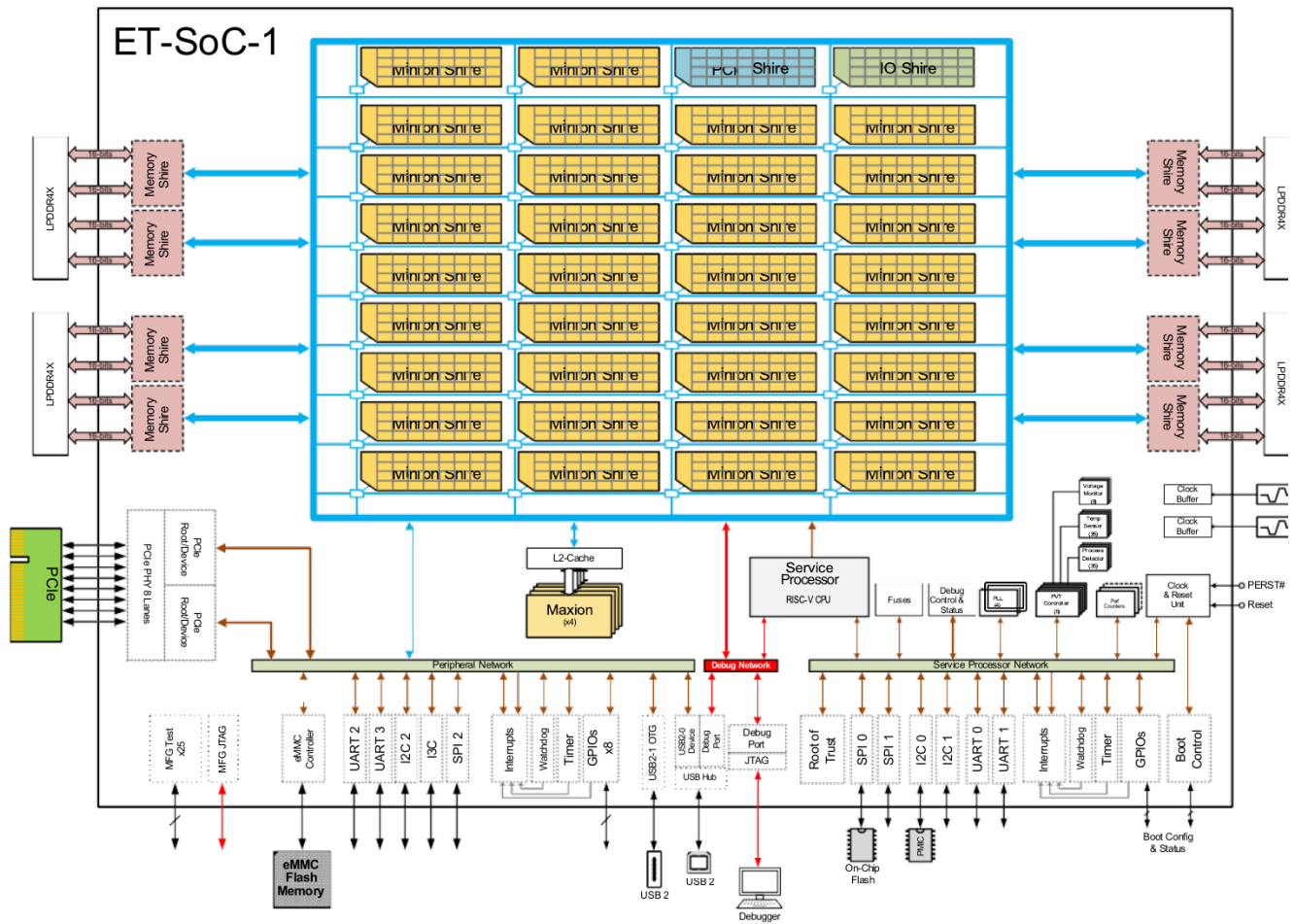


Figure 6. ET-SoC-1 block diagram

1.4.1. Mesh Topology and Mesh Stops

The mesh is organized as an 8×6 grid, and each Shire attaches to it at a mesh stop. The Memory Shires occupy the east and west edges, four on each side, which leaves the corners of the grid unoccupied. The result is 44 mesh stops, listed in [Table 2](#).

Table 2. Shires on the mesh

Shire	Count	Contents
Minion Shire	34	32 ET-Minion cores and a 4 MB Shire Cache
Memory Shire	8	LPDDR4X memory controller
PCIe Shire	1	PCIe controllers and PHY
I/O Shire	1	ET-Maxions, Service Processor and peripherals
Total	44	

1.4.2. Memory Shires

Each Memory Shire contains a 32-bit LPDDR4X PHY and two memory controllers, each driving one 16-bit channel achieving 4,266 MT/s/channel. The Memory Shires also support global atomic operations.

1.4.3. PCIe Shire

The PCIe Shire contains two dual-mode PCIe Gen4 controllers that share an 8-lane PHY. It supports three lane configurations:

- Single 8-lane endpoint or root complex
- Single 4-lane endpoint or root complex
- 4-lane endpoint together with a 4-lane root complex

The interface is normally factory-configured as an endpoint.

1.4.4. I/O Shire

The I/O Shire contains the ET-Maxions and the Service Processor ([Section 1.3](#)). It also contains the device peripherals, which are grouped on three networks according to which agents may access them.

- **Service Processor network.** Accessible only to the SP. It contains:
 - SPI, including the boot-flash controller
 - I2C, including the link to the external power-management controller
 - UART, including the SP console
 - GPIO, timers, watchdog and boot control
- **Peripheral network.** Shared by the SP, the ET-Maxions and the master Minion Shire. It contains:
 - eMMC, USB 2.0 OTG, I2C/SMBus, I3C, SPI and UART
 - GPIO, timers and watchdog
- **Debug network.** Provides debug access through JTAG and a USB 2.0 device port.

1.5. Memory Map

1.6. ET-SoC-1 Accelerator Card

ET-SoC-1 operates as an accelerator attached to a host system over PCI Express. It is mounted on a PCIe add-in card (Figure 7) with the following features:

- PCIe x16 card-edge connector, with an 8-lane Gen4 interface to ET-SoC-1
- ET-SoC-1 soldered to the card
- Four LPDDR4X devices forming a 256-bit memory interface
- 64 GB eMMC flash memory
- USB connectors for the ET-SoC-1 device and debug ports and for a UART interface, and a JTAG connector
- DIP switches that select the boot options

Multiple ET-SoC-1 devices can also be integrated on a single card.



Figure 7. ET-SoC-1 PCIe card

2. ET-Minion Core

2.1. Core Overview

This section describes the internal organization of the ET-Minion core: its block partitioning, hardware threads and instruction flow. The features of the core and its position in the ET-Minion hierarchy are summarized in [Section 1.2.1](#).

The ET-Minion is a dual-threaded, in-order, single-issue 64-bit RISC-V core that implements RV64IMFC with the Esperanto instruction extensions. It is implemented in `minion_top` and consists of four units: the Front End, the integer pipeline, the L1 data cache and the VPU. The core fetches instructions from the Neighborhood L0 micro-cache and accesses memory through the Neighborhood memory path; it has no instruction cache of its own.

2.1.1. Block Partitioning

The ET-Minion core is implemented in `minion_top` ([Figure 2](#)). `minion_top` contains two blocks: `core_top`, which holds the Front End, the integer pipeline, the L1 data cache and the debug APB slave, and `vpu_top`, the vector processing unit. [Table 3](#) lists the blocks.

Table 3. ET-Minion core blocks

Block	Module	Function
Front End	<code>frontend_top</code>	Fetches instructions for both threads from the Neighborhood L0 micro-cache, expands compressed instructions, decodes them for the integer pipeline and the VPU, and issues one instruction per cycle to the integer pipeline (Section 2.2).
Integer pipeline	<code>intpipe_top</code>	In-order, single-issue pipeline with the stages ID, EX, GSC, TAG, MEM and WB. Contains the integer register file, ALU, multiply/divide unit, integer, FP and mask scoreboards, and the CSRs. Redirects the Front End on a change of control flow.
Data cache	<code>dcache_top</code>	4 KB private L1 data cache, partly configurable as scratchpad memory. Contains the data TLB, PMA check, miss handling, replay queue, store merge, atomic (AMO) forwarding to L2 through the miss handler's UC flow, scratchpad controller, write-back unit, cache management, tensor-load and reduce units, and the interface to the Neighborhood memory path.
APB slave for debug registers	<code>minion_debug_apb_slv</code>	APB slave for debug access to the core, including writes to the debug program buffer.
VPU	<code>vpu_top</code>	Vector processing unit with a control block and eight identical lanes. Executes floating-point, packed-integer, transcendental and tensor operations.

2.1.2. Hardware Threads

The core runs two hardware threads (`CORE_NR_THREADS` = 2). Each thread has its own architectural state; the pipeline and execution units are shared, and instructions of the two threads are interleaved in the pipeline.

Table 4. Per-thread and shared resources

Per thread	Shared by both threads
Fetch PC, thread buffer and instruction FIFO in the Front End	Front End request port, decoders and thread scheduler
32 x 64-bit integer registers, in one register file indexed by thread	Integer pipeline, ALU and multiply/divide unit
32 vector registers, (also the FP f-registers), 256 bits wide, held as a 32-bit slice in each lane's RF, which are indexed by the thread number. Eight 8-bit mask registers.	VPU control and execution units
CSR state, interrupt inputs and debug state	L1 data cache and scratchpad

Each thread is enabled by its corresponding bit of the `enabled` input of `minion_top`. `minion_top` registers `enabled` once, and the integer pipeline and debug logic uses this registered copy directly. The Front End registers it twice more, so that it can detect the edges of the enable and turn each rising and falling edge into a one-cycle enable of disable event ([Section 2.2.6.3](#)). When a thread is enabled, it starts fetching at `reset_vector`, which is the same for both threads.

The Front End selects one thread per cycle for issue to the integer pipeline. The integer pipeline can stall each thread independently, for example on `WFI`, a fence, a scoreboard hazard, or while the other thread holds exclusive mode. As a result, the Front End sees a change of `enabled` two cycles later than the integer pipeline does.

2.1.3. Instruction Flow Through the Core

The Front End fetches instructions for both threads from the Neighborhood L0 micro-cache, expands compressed instructions, decodes them, and presents one instruction per cycle to ID, the first stage of the integer pipeline. The integer pipeline executes the instructions in order and passes memory operations to the data cache and vector operations to the VPU. In the opposite direction, the integer pipeline redirects the Front End to a new fetch address after a branch misprediction, exception or other change of control flow, and stalls a thread that must not issue.

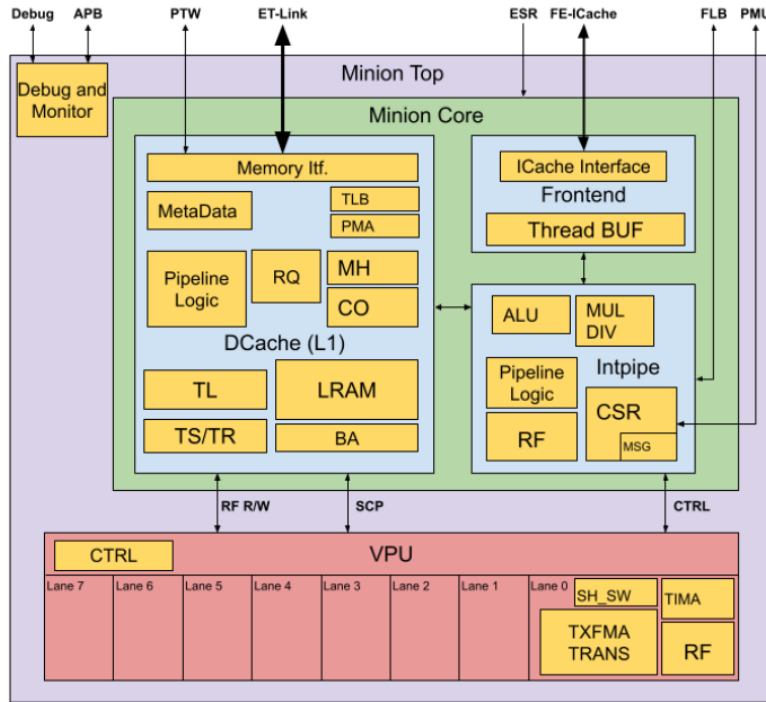


Figure 8. Front End in the ET-Minion core

2.1.4. Interfaces to the Neighborhood

Table: ET-Minion core interface groups.

2.1.5. Clocks, Resets and Power

2.1.6. Debug Support

2.2. Front End

2.2.1. Overview

The Front End (`frontend_top`) supplies instructions to the ET-Minion core. It sits between the Neighborhood L0 micro-cache and the ID stage of the integer pipeline. It holds the program counter of each of the two threads and fetches 32-byte instruction blocks from the L0 micro-cache. It then extracts the instruction, expands compressed instructions to 32 bits and decodes them for both the integer pipeline and the VPU. It delivers one decoded instruction per cycle to ID, from either thread. The core has no instruction cache of its own. Each L0 micro-cache is shared by four ET-Minion cores, eight threads in all, so the design of the Front End is driven by the need to hide the latency of shared fetch path.

The Front End is built around the following four concepts:

1. **Fetch is double-buffered:** each thread has a thread buffer with two 256-bit entries, each holding half a cache line, so the next block can be requested while the current one is being consumed.
2. **Each thread has at most one fetch request in flight:** the L0 micro-cache answers every accepted request, hit or miss, after a fixed latency, so a thread recognizes its own by timing

alone, without a thread identifier. On a miss the thread sleeps until the L0 micro-cache reports that a fill has completed, and then retries.

3. **Instructions are decoded before they are queued:** one integer decoder and one VPU decoder, shared by both threads, decode in F7, and each thread has a two-entry instruction FIFO that holds decoded instructions until ID accepts them. This moves decode out of the ID stage and lets a thread keep fetching while ID serves the other thread.
4. **The two threads are independent:** each thread fetches, sleeps, stalls and restarts on its own. They share only the fetch request port, the decoders and the issue slot into ID, and the thread scheduler alternates between them when both are eligible.

The Front End is a pipeline of eight stages, F0 to F7. F1 to F5 overlap the L0 micro-cache pipeline. With an L0 hit and no competing requests, the first instruction of a block reaches ID nine cycles after it is requested. The first target instruction reaches ID ten cycles after a redirect, and the first instruction reaches ID eleven cycles after a thread is enabled.

The integer pipeline controls the Front End through two per-thread signals. A redirect (`f0_core_req_valid`, `f0_core_req`) flushes the thread and restarts fetch at a new PC. It is used for mispredictions, exceptions, replays, interrupts and debug-mode transitions. A stall (`id_core_stall`) stops the thread from being scheduled, for example on WFI, a fence, a CSR access that must wait, or a scoreboard hazard. The Front End attaches the fetch status to each instruction: page and access faults, bus and ECC errors, and a replay flag for uncacheable fetches issued while older instructions are still in flight. The integer pipeline acts on this status in ID.

The Front End also starts and stops threads through their enable inputs and reset vector. It supports debug in two ways: it generates the placeholder instruction on which a halt is taken, and it lets entry 0 of a halted thread's buffer serve as the debug program buffer. Each thread buffer runs on its own gated clock; a chicken bit disables the gating.

Table 5 lists the submodules and Figure 9 shows how they are connected. The sections that follow cover the interface (Section 2.2.2), the pipeline stages (Section 2.2.3), the fetch protocol with the L0 micro-cache (Section 2.2.4), the thread buffer (Section 2.2.5 and Section 2.2.6), decode (Section 2.2.7), the instruction FIFO (Section 2.2.8), the thread scheduler (Section 2.2.9), and halt and debug support (Section 2.2.10).

Table 5. Front End submodules

Submodule	Instances	Function
Thread buffer (<code>frontend_thread_buffer</code>)	2, one per thread	64-byte instruction buffer and the thread's program counter. Issues fetch requests to the L0 micro-cache.
RVC expander (<code>frontend_rvc_expander</code>)	2, one per thread buffer	Expands compressed (RVC) instructions to 32 bits.
Request arbiter (<code>arb_lru_data</code>)	1	2:1 LRU arbiter. Selects the thread whose fetch request is sent to the L0 micro-cache.
Integer decoder (<code>intpipe_decode</code>)	1	Decodes the instruction into integer-pipeline control signals.

Submodule	Instances	Function
VPU decoder (<code>vpu_decoder</code>)	1	Decodes the instruction into VPU control signals.
Instruction FIFO (inline generate block)	2, one per thread	Two-entry queue of decoded instructions waiting for the integer pipeline.
Thread scheduler (<code>frontend_thread_sched</code>)	1	Selects the thread whose instruction is presented to the integer pipeline.

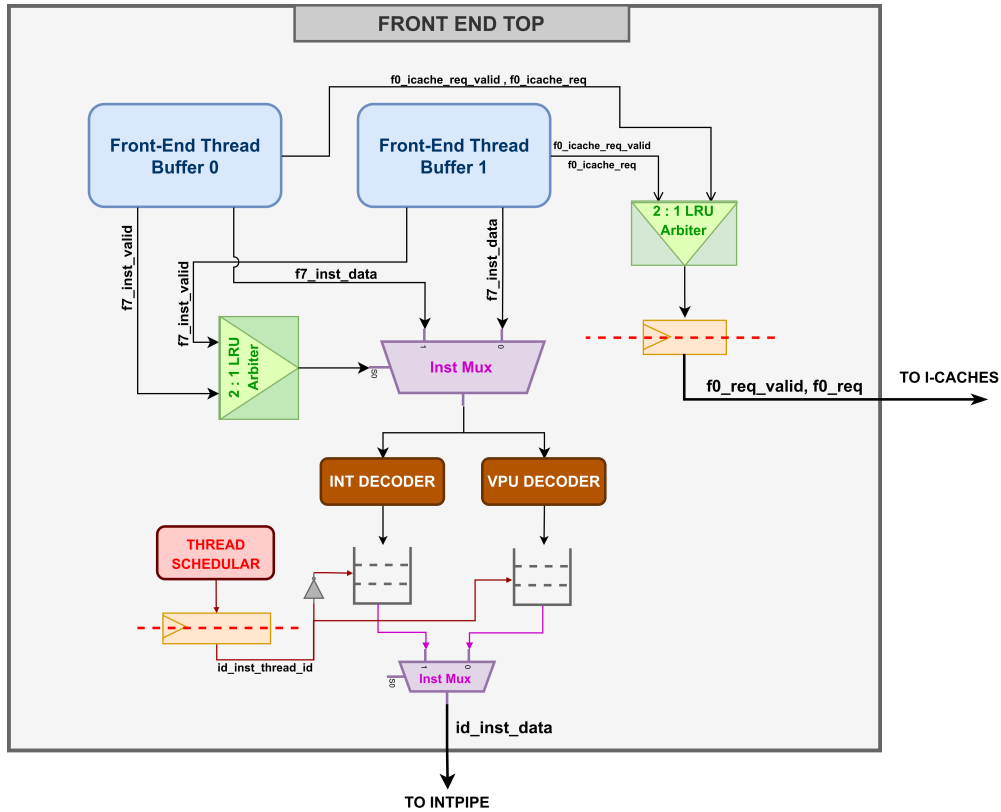


Figure 9. Front End block diagram

2.2.2. Interface

This section defines the boundary of `frontend_top`. Section 2.2.2.1 lists its ports in six groups. The system group has the clock, the warm and debug resets and the clock-gating chicken bit. Thread control has the thread enables, the reset vector and the virtual-memory status of each thread. The L0 micro-cache group has the fetch request and response and the fill-done signal. The integer-pipeline group has the per-thread redirect and stall inputs. The output group carries the instruction and it's decoded data to ID. The debug group has the halt, halted and program-buffer signals. For each port the table gives the net it connects to in `core_top`.

Section 2.2.2.2 gives the field layout and bit width of each packed structure that crosses these ports. These are the fetch request and response, the virtual-memory status, the redirect, the instruction delivered to ID, the instruction FIFO entry, and the two decoded control words. Section 2.2.2.3 lists the defines that set the number of threads, the buffer geometry and the PC width.

2.2.2.1. I/O Signals

Table 6 lists the ports of `frontend_top`. *Connected to* gives the signal in `core_top`. A width of 2 denotes one bit per thread.

Table 6. Front End I/O signals

Signal	Dir	Width / type	Connected to	Description
System				
<code>clock</code>	in	1	<code>clock</code>	Core clock. Each thread buffer derives a gated clock from it.
<code>reset</code>	in	1	<code>reset_w</code>	Warm reset. Synchronous.
<code>reset_debug</code>	in	1	<code>reset_d</code>	Debug reset. Clears the program-buffer control and, while the thread is halted, refills the program buffer with <code>EBREAK</code> .
<code>chicken_bit</code>	in	1	<code>chicken_bit_frontend</code>	Disables Front End clock gating. Driven by the <code>min_frontend_clock_gate_disable</code> chicken bit.
Thread control				
<code>f0_thread_enabled</code>	in	2	<code>enabled</code>	Thread enable. A rising edge starts fetch at the reset vector; a falling edge stops fetch.
<code>f0_reset_vector</code>	in	48	<code>reset_vector</code>	Fetch address after reset or thread enable.
<code>vm_status</code>	in	2 x <code>minion_vm_status</code>	<code>vm_status</code>	Virtual-memory status of each thread, forwarded with every fetch request.
Fetch request to the L0 micro-cache				
<code>f1_icache_req_valid</code>	out	1	<code>icache_req_valid</code>	The F1 I-cache request register holds a request.
<code>f1_icache_req</code>	out	<code>frontend_icache_req</code>	<code>icache_req</code>	Fetch request: thread, VM status and address.
<code>f1_icache_req_ready</code>	in	1	<code>icache_req_ready</code>	The Neighborhood accepts the request in this cycle.
Fetch response from the L0 micro-cache				
<code>f5_icache_resp_valid</code>	in	1	<code>icache_resp_valid</code>	A response is present. It carries no thread identifier.

Signal	Dir	Width / type	Connected to	Description
f5_icache_resp_miss	in	1	icache_resp_miss	The response is a miss and carries no instruction data.
f5_icache_resp	in	icache_frontend_resp	icache_resp	32-byte block and status.
f6_icache_fill_done	in	1	icache_fill_done	The L0 micro-cache has completed a line or translation fill.
Redirect and stall from the integer pipeline				
f0_core_req_valid	in	2	id_core_fe_req_valid	Redirect. Flushes the thread and restarts fetch at f0_core_req.pc.
f0_core_req	in	2 x minion_fe_req	id_core_fe_req	Redirect PC, speculation status and debug-mode transition.
id_core_stall	in	2	id_core_fe_stall	Thread must not issue. Raised for WFI, CSR accesses that are replayed until a resource is ready (e.g., TensorWait, tensor ops, FCC), a fence, exclusive mode held by the other thread, and integer or FP scoreboard stalls. The WFI/CSR term is suppressed while debug mode or halt request is active.
Instruction to the integer pipeline (ID)				
id_inst_valid	out	1	id_fe_core_resp_valid	The instruction presented to the integer pipeline is valid.
id_inst_ready	in	1	id_core_fe_resp_ready	The integer pipeline is ready to accept the instruction.
id_inst_thread_id	out	1	id_fe_core_resp_thread_id	Thread of the instruction.
id_inst_data	out	frontend_core_resp	id_fe_core_resp	Instruction, PC and fetch status.
id_intpipe_ctrl	out	minion_control	id_intpipe_ctrl	Integer decode.
id_vpu_decoder_sigs	out	vpu_ctrl_sigs_t	id_vpu_decoder_sigs	VPU decode. Valid only for VPU instructions.

Signal	Dir	Width / type	Connected to	Description
id_vpu_core_ctrl	out	vpu_minion_id_ctrl	id_frontend_vpu_ctrl	VPU and mask register usage, for hazard checks in the integer pipeline.
Debug				
halt	in	2	debug_in_core.halt	A halt request is pending.
halted	in	2	debug_out.halted	The thread is in debug mode.
debug_ffb_wdata	in	64	debug_ffb_wdata	Program-buffer write data.
debug_ffb_en	in	4	debug_ffb_en	Program-buffer word write enables.
debug_ffb_thread_sel	in	1	debug_ffb_thread_sel	Thread whose program buffer is written.
debug_ffb_exec	in	2	debug_ex_program_buffer	Start executing the program buffer.

2.2.2.2. Data Structures

Table 7. `frontend_icache_req`: fetch request (58 bits)

Field	Width	Description
thread_id	1	Requesting thread.
vm_status	minion_vm_status	Virtual-memory status of the thread (Table 8).
addr	49	Fetch address. The L0 micro-cache returns the aligned 32-byte block that contains it.

Front End addresses are 49 bits wide: the 48-bit virtual address and one additional bit that allows a non-canonical 64-bit address to be represented.

Table 8. `minion_vm_status`: virtual-memory status (8 bits)

Field	Width	Description
prv	2	Current privilege level.
mprv, mpp, sum, mxr	1, 2, 1, 1	Corresponding <code>mstatus</code> fields.
debug	1	Thread is in debug mode.

Table 9. `icache_frontend_resp`: fetch response (261 bits)

Field	Width	Description
data	256	32-byte instruction block.
page_fault	1	Instruction page fault.
access_fault	1	Instruction access fault.

Field	Width	Description
cacheable	1	Block is in a cacheable region.
bus_err	1	Bus error on the line.
ecc_err	1	ECC error on the line.

Table 10. `minion_fe_req`: redirect (52 bits)

Field	Width	Description
pc	49	New fetch PC.
speculative	1	Thread has older instructions in the integer pipeline. Driven every cycle, independently of <code>f0_core_req_valid</code> . Used for replay (Section 2.2.5.6).
debug_info.halt	1	Redirect enters debug mode.
debug_info.resume	1	Redirect leaves debug mode.

Table 11. `frontend_core_resp`: instruction to the integer pipeline (89 bits)

Field	Width	Description
pc	49	PC of the instruction.
inst_bits	32	Instruction. Compressed instructions are delivered expanded.
pf0, af0	1 each	Page fault / access fault on the entry holding the start of the instruction.
pf1, af1	1 each	Page fault / access fault on the entry holding the upper half of an instruction that crosses entries.
replay	1	Instruction must be replayed.
rvc	1	Instruction was compressed.
bus_error, ecc_error	1 each	Bus error / ECC error on the entry or entries holding the instruction.

Table 12. `frontend_thread_data`: instruction FIFO entry (179 bits)

Field	Width	Description
core_resp	89	Instruction and fetch status (Table 11).
vpu_ctrl_sigs	45	VPU decode (<code>vpu_ctrl_sigs_t</code>).
intpipe_ctrl	45	Integer decode (<code>minion_control</code>).

Table 13. Decoded control words

Structure	Width	Content
<code>minion_control</code>	45	Legal instruction; software-emulated (M-code) instruction; branch, <code>JAL</code> , <code>JALR</code> ; integer source reads and destination write; ALU operand selects, function and width; immediate format; memory operation, command and size; FP/VPU destination write; mask register read/write; multiply/divide; CSR command; fence; global memory access; gather/scatter; graphics; implicit use of <code>x31</code> .
<code>vpu_ctrl_sigs_t</code>	45	VPU function; execution unit (TxFMA, shift/swizzle, transcendental ROM); conversion and load/store type; source, third-source and destination reads and write; mask register reads; operand swaps; scalar or packed; data type; integer-register transfer; rounding; FP flag update.

2.2.2.3. Configuration Parameters

Table 14. Front End configuration parameters

Define	Value	Description
<code>CORE_NR_THREADS</code>	2	Threads per core.
<code>FE_FETCH_BUFFERS</code>	2	Thread-buffer entries per thread.
<code>FE_FETCH_READ_SIZE</code>	256	Bits per fetch block and per entry.
<code>FE_FETCH_PTR_SIZE</code>	5	Read-pointer width: 1 entry bit and 4 halfword bits. $\log_2(\text{FE_FETCH_VALID_INST_SIZE} * \text{FE_FETCH_BUFFERS}) = \log_2(16 * 2) = 5$
<code>PC_SIZE_EXT</code>	49	Front End PC width.

2.2.3. Pipeline Stages

This section describes the eight Front End stages, F0 to F7, and how they line up with the L0 micro-cache pipeline. Two activities run in parallel and meet in the thread buffer. Fetch runs from F0 to F6: a thread's request is arbitrated and sent in F0, waits for acceptance in F1, is looked up in F1 to F5, and the response is written into the thread buffer in F6; a miss writes nothing and the thread goes to sleep. In F7 each thread pulls its next instruction out of its buffer and expands it if compressed. One thread's instruction is decoded and placed in its FIFO. The scheduler presents one thread's FIFO head to the integer pipeline each cycle

Fetching and extracting are independent: a thread keeps extracting instructions from a filled buffer entry while its next request is still in F0-F5.

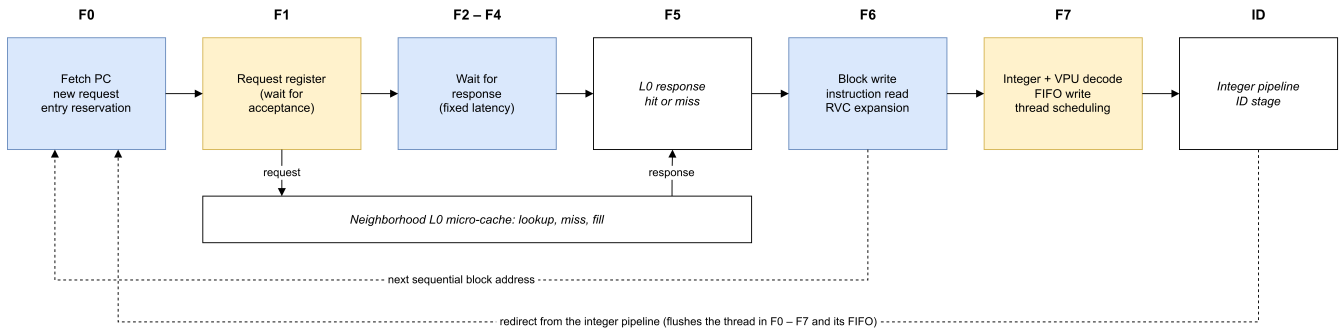


Figure 10. Front End pipeline

F0 Stage

The F0 stage receives instruction-block requests from both thread buffers and decides which one gets access to the L0 micro-cache using a 2:1 LRU arbiter. The selected request is registered and sent to the L0 micro-cache through the Neighborhood interface. This stage also handles the enabling and disabling of threads and redirect requests from the integer pipeline, for example the new address after a mispredicted branch.

F1–F5 Stage

The selected request is held in F1 until the L0 micro-cache accepts it, which happens at the earliest in the second F1 cycle. The thread then waits for the response. The L0 micro-cache always responds to an accepted request, whether it is a hit or a miss, with a fixed five-cycle latency: counting the cycle in which it accepts the request as the first cycle, the response is returned in the fifth cycle, in F5 (Figure 12).

F6 Stage

The response from the L0 micro-cache is checked. If it is a hit, the instruction block is stored in the buffer of the thread that issued the request. If it is a miss, nothing is stored, and the thread waits for the cache to be filled. Instruction reading is independent of the fetch request: whenever a buffer entry holds instructions, the thread reads its next instruction, expands it if it is compressed, and passes it on for decoding, in parallel with any request in F0 to F5.

F7 Stage

This stage selects which thread's instruction is read from its buffer. The selected instruction is decoded and stored, along with its control signals, in that thread's FIFO. The thread scheduler decides which thread's FIFO issues an instruction to the integer pipeline in the next cycle. If both threads are eligible, they are selected in round-robin order; otherwise, the eligible thread is selected.

With an L0 hit and no competing requests, the first instruction of a block reaches ID nine cycles after its request is issued in F0.

2.2.4. Interaction with the L0 Micro-Cache

This section describes the fetch protocol between the Front End and the L0 micro-cache, from the Front End side. It covers the following four things:

1. How a request is arbitrated and accepted: by the 2:1 LRU arbiter and F1 request register in the

core, then by the 8:1 arbiter of the Neighborhood.

2. The fixed response timing, and how a thread recognizes its own responses with a token tracker, since responses carry no thread identifier.
3. The miss, sleep and fill-done protocol, and why its timing guarantees that no wake-up is lost.
4. How a redirect cancels requests already in flight, including one still waiting in the F1 register.

Each L0 micro-cache serves the eight threads of four ET-Minion cores. [Figure 11](#) shows the request and response path of one core. The Neighborhood side of the interface, drawn in grey, is described in [Section 3.3](#) and in the sections that follow it.

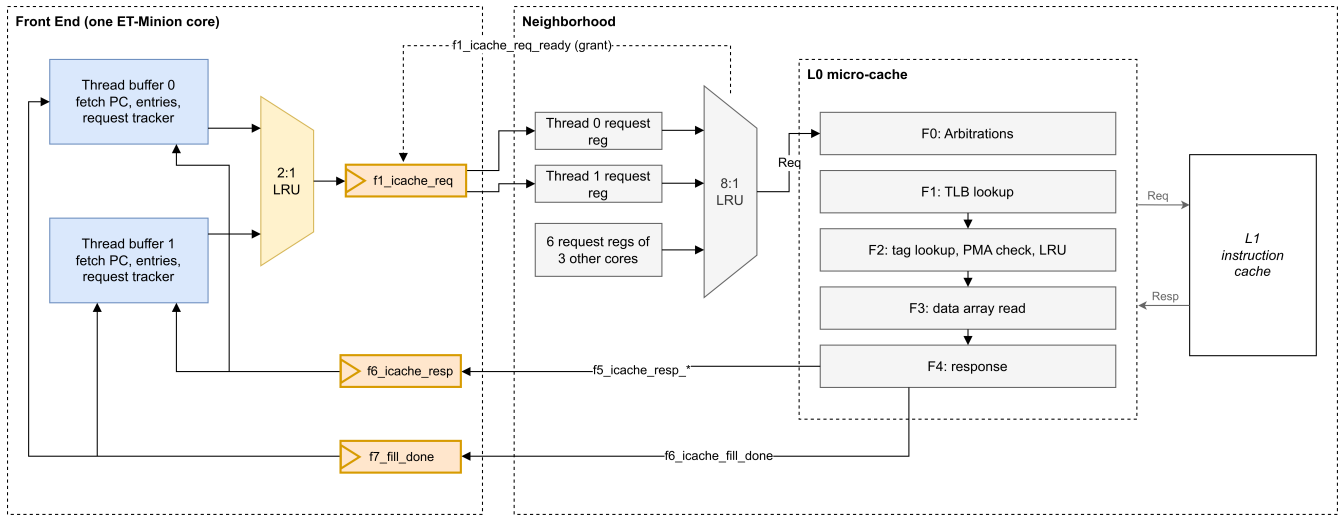


Figure 11. Front End to L0 micro-cache request and response path

2.2.4.1. Request Arbitration and Acceptance

A fetch request passes through the following steps:

1. **2:1 LRU arbiter, F0.** Selects between the two threads of the core. With two clients the grant alternates when both threads request. The priority is updated on every grant.
2. **F1 I-cache request register (f1_icache_req).** Holds the granted request until the Neighborhood accepts it with **f1_icache_req_ready**. The register accepts a new request when it is empty or its current request is accepted in the same cycle; otherwise the 2:1 arbiter issues no grant.
3. **8:1 LRU arbiter, Neighborhood.** The request is first captured in a per-thread request register, then arbitrated against the other seven threads served by the same L0 micro-cache. The grant of this arbiter is returned as **f1_icache_req_ready**. Because of the capture register, a request is accepted at the earliest in its second F1 cycle.

2.2.4.2. Response Timing and Ownership

The L0 micro-cache has a five-stage pipeline and returns exactly one response per accepted request, from its last stage, for a hit and for a miss alike. The response reaches the Front End in F5.

The response carries no thread identifier. Each thread buffer identifies its responses by position in time: when its request is granted, the thread inserts a token (a valid bit and the reserved entry number) into its own F1–F6 tracker. The token remains in F1 until the request is accepted and then advances one stage per cycle. A response registered in F6 together with a valid token belongs to that

thread, and the token selects the entry to be written. The response and fill-done signals are registered in `frontend_top` and broadcast to both thread buffers.

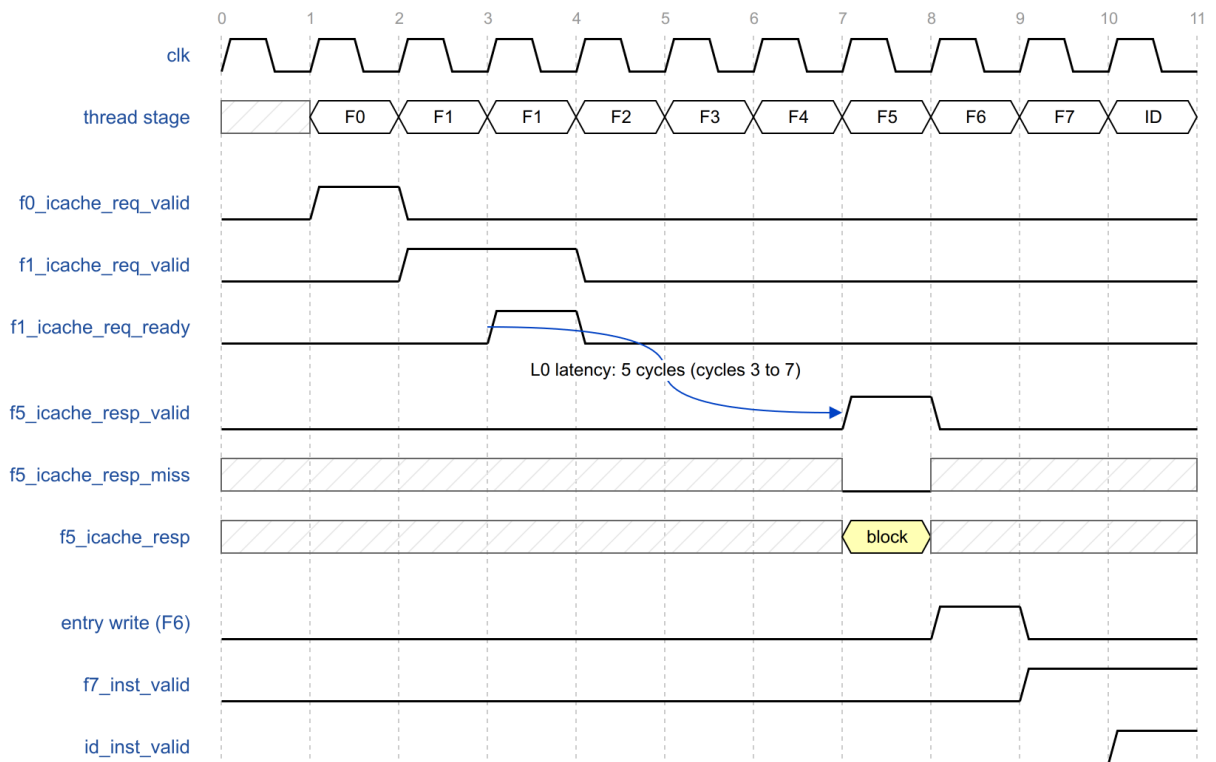


Figure 12. Request and hit response

In Figure 12 the request is issued in cycle 1, accepted in cycle 3, and the block returns in cycle 7. It is written in cycle 8; the first instruction of the fetched block is in the F7 stage in cycle 9 and in the ID stage in cycle 10.

2.2.4.3. Miss Handling and Fill Done

A miss response has the same timing as a hit and carries no data. On a miss the thread writes nothing, keeps its fetch PC and enters a sleep state (`f0_miss_pending`) in which it issues no requests.

On completion of a line fill from the L1 instruction cache, or of a translation fill, the L0 micro-cache pulses `f6_icache_fill_done`. The signal is broadcast to all threads of the L0 micro-cache and carries no address. Every sleeping thread wakes and retries; a thread whose line is still absent misses again and sleeps until the next fill. A redirect, a thread disable or a pending halt also clears the sleep state.

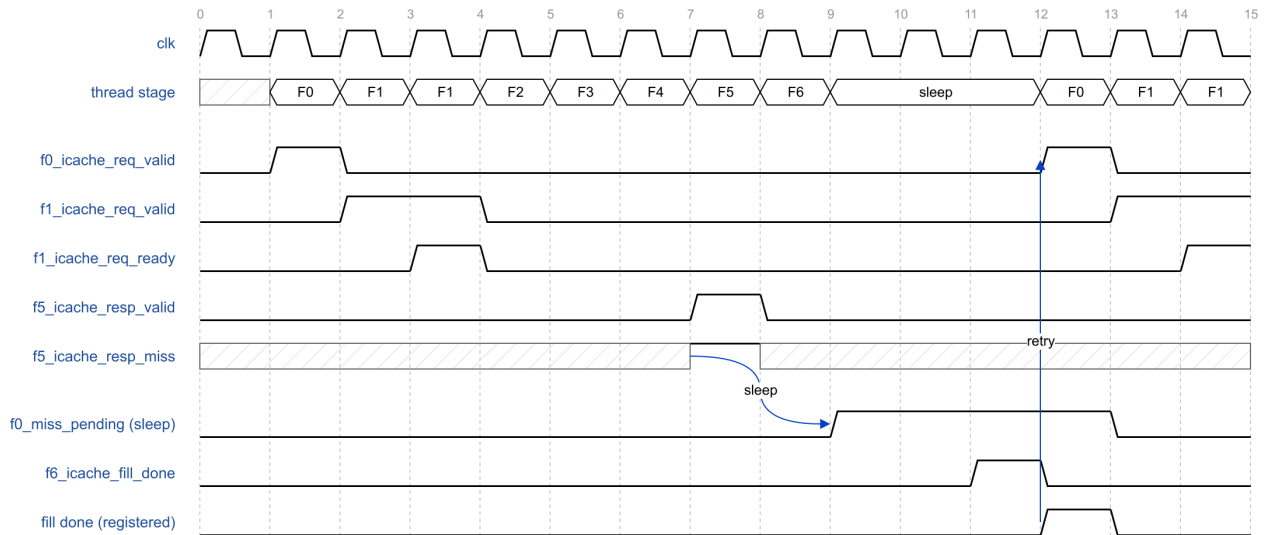


Figure 13. Miss, sleep and retry

In Figure 13 the miss response arrives in cycle 7 and the thread sleeps from cycle 9. Fill done arrives in cycle 11, is registered, and the thread retries in cycle 12.

Fill done is issued one cycle after the response slot. A request that looks up the L0 micro-cache while a fill is being written still misses, and its response is the last miss that the fill can cause. Issuing fill done one cycle after that response, and registering it again in the Front End, ensures that every thread has entered the sleep state before the wake-up arrives. A wake-up that arrived first would be lost, and the thread would wait for an unrelated fill.

2.2.4.4. Request Cancellation

A redirect invalidates the thread's tokens in F2 to F5 and suppresses the F6 write, so the responses of in-flight requests are discarded.

A request still waiting in the F1 I-cache request register cannot be withdrawn: the register is shared and the Neighborhood already holds a copy. The thread records the cancellation and does not advance the token when the request is accepted. The I-cache lookup is performed and its response is discarded. The thread's new request is issued only after the cancelled request has been accepted.

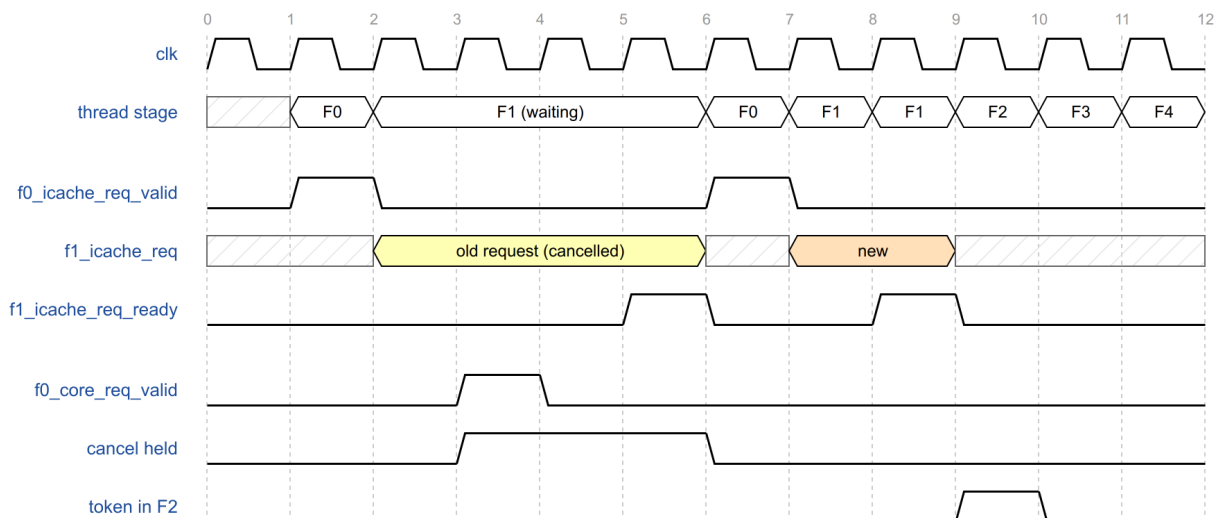


Figure 14. Redirect while a request waits in F1

In Figure 14 the redirect arrives in cycle 3 while the request waits in F1. The cancellation is held until the request is accepted in cycle 5; no token enters F2. The new request is issued in cycle 6 and accepted in cycle 8.

2.2.5. Thread Buffer

The thread buffer (`frontend_thread_buffer`) holds almost all of a thread's Front End state; there is one buffer per thread. This section describes how the buffer stores instructions: two 256-bit entries in a latch register file with a 32-bit, wrap-around read port. It then describes the state kept for each entry, the fetch PC and how it is updated, and the conditions under which the thread requests a new block. It also shows how an entry is reserved and written.

The section then shows the instruction side: how the read pointer extracts one instruction per cycle, including instructions that cross from one entry to the other and the bypass from incoming data. It then covers how the RVC expander turns compressed instructions into 32-bit ones, and which fault, error and replay status is attached to each instruction on its way to the integer pipeline.

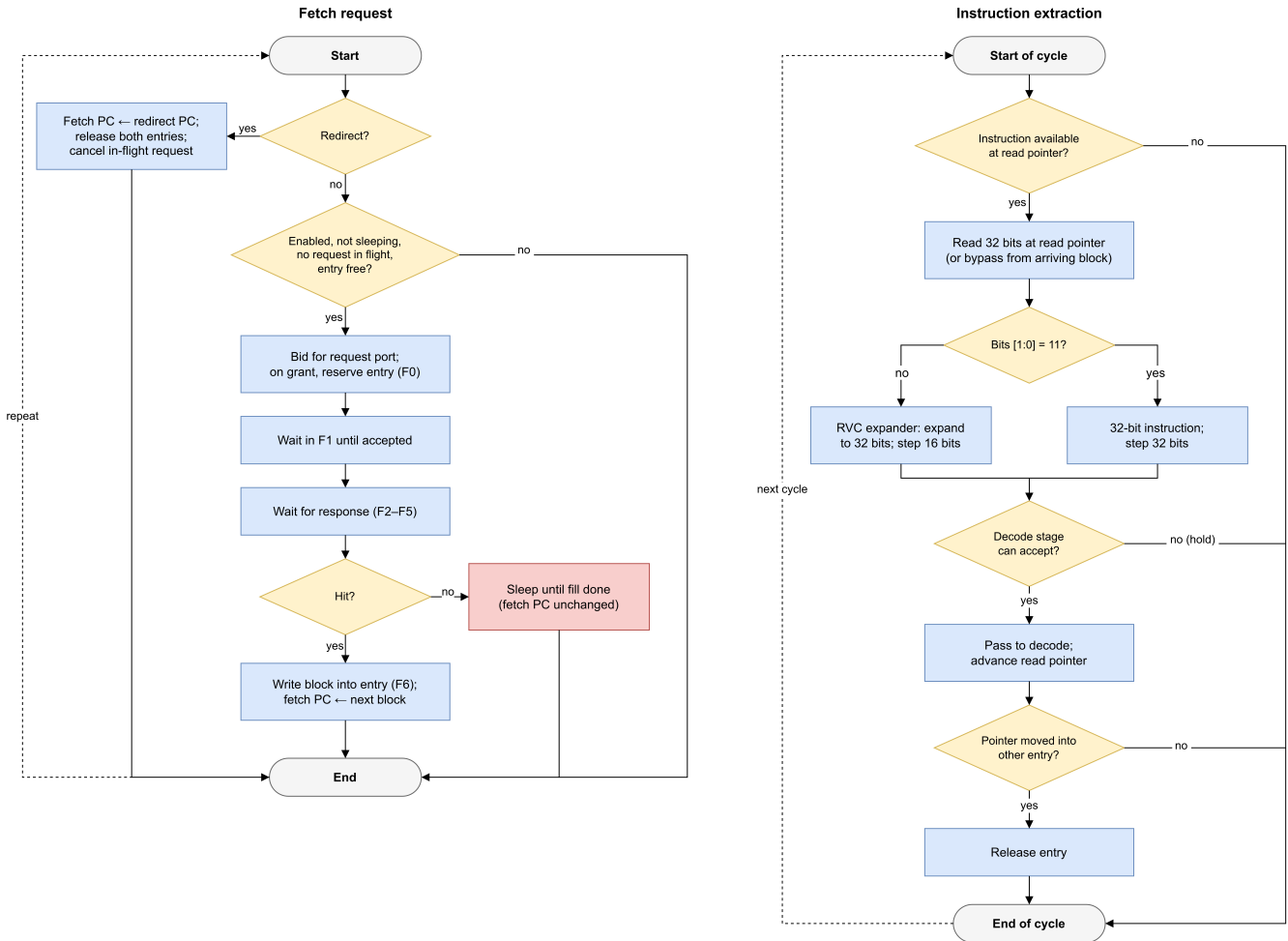


Figure 15. Thread buffer operation

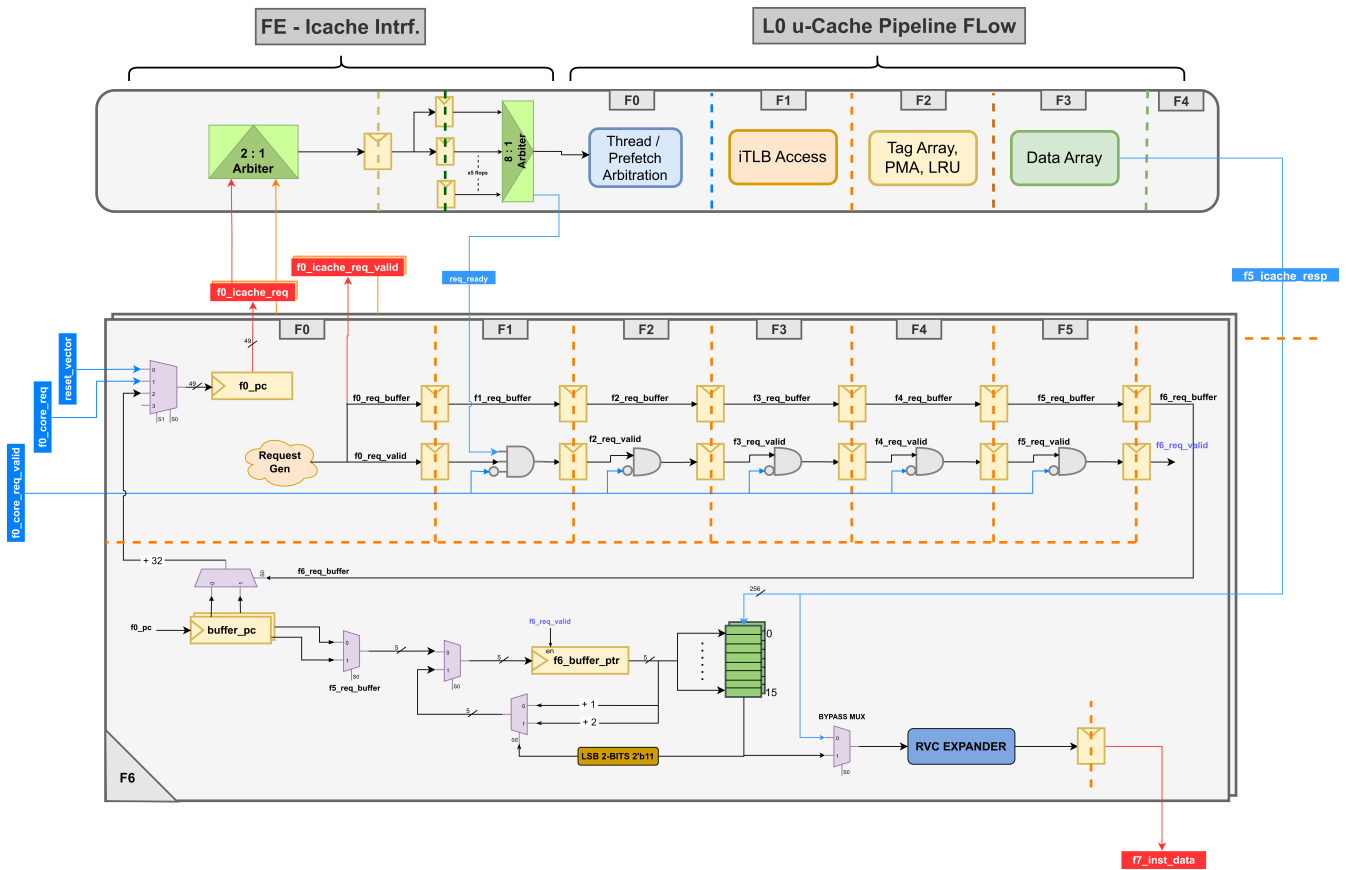


Figure 16. Thread buffer pipeline

2.2.5.1. Buffer Entries

The instruction buffer has two entries of 256 bits (32 bytes). An entry holds one fetch block: eight 32-bit instructions, sixteen compressed instructions, or a combination. Two entries allow the next block to be fetched while the current block is consumed.

The entries are implemented as one latch-based register file with two ports:

- **Write port, 256 bits.** Writes a complete block into one entry, with one write enable per 32-bit word.
- **Read port, 32 bits.** Addressed by the 5-bit read pointer `buffer_ptr`: bit 4 selects the entry, bits 3:0 select a halfword within it. The read returns the 32 bits starting at that halfword.

The read port wraps around: halfword 15 of one entry is followed by halfword 0 of the other. The entries therefore form a ring of 32 halfwords, and a 32-bit instruction that starts in the last halfword of an entry is read in one access.

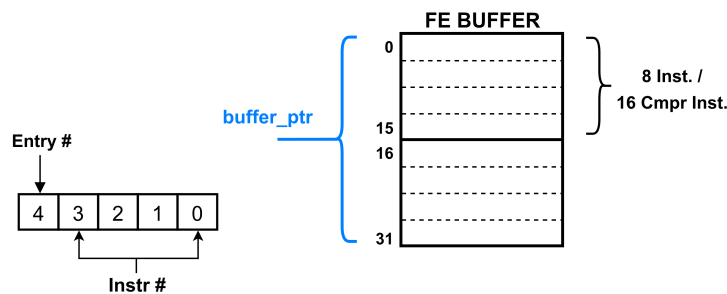


Figure 17. Thread-buffer entries and read pointer

Table 15. Thread-buffer entry state

State	Description
Empty flag	Entry holds no instruction still to be sent. Set for both entries at reset.
Block address	Address of the instruction block held in the entry (<code>buffer_pc</code>). Only the words from this address onward are written, extraction in the entry starts at this address, and it provides the upper PC bits of every instruction read from the entry.
Fault flags	Page fault and access fault of the block.
Error flags	Bus error and ECC error of the block.
Cacheable flag	Block is in a cacheable region. Used for replay.

2.2.5.2. Program Counter

Each thread buffer holds a 49-bit fetch PC, the address of the next block to request. It is updated as listed in [Table 16](#).

Table 16. Fetch PC update

Event	New fetch PC
Reset, or thread enabled	<code>f0_reset_vector</code> , sign-extended.
Block written into an entry	Start of the next aligned 32-byte block.
Redirect (highest priority)	<code>f0_core_req.pc</code> .

Only the first block after a reset or redirect can start within a 32-byte block; all later blocks are aligned. A miss does not change the fetch PC, so the same address is requested again after the fill.

The PC of each extracted instruction is formed from the upper bits of its entry's block address and the halfword position of the read pointer.

2.2.5.3. Request Generation and Entry Fill

A thread issues a request in F0 when all conditions in [Table 17](#) are met.

Table 17. Request conditions

Condition	Reason
Thread enabled, and not in the enable cycle	Reset vector is loaded at the end of the enable cycle.
No redirect in the current cycle	Redirect PC is loaded at the end of the cycle.
Thread not sleeping, or registered fill done in this cycle	After a miss the thread waits for a fill; the request is issued in the cycle the registered fill done arrives.
No request of the thread in F1 to F6	At most one request per thread is in flight, which keeps responses in order and allows ownership by timing.

Condition	Reason
An entry is empty or is released in the current cycle	The response cannot be stalled and must have a destination.
No halt pending	While a halt is pending the thread sends no request to the L0 micro-cache; it produces a placeholder instruction instead (Section 2.2.10.1).
Not in reset; not executing the debug program buffer	—

On a grant, the thread reserves an entry and records the fetch PC of the request as the entry's block address. If both entries are empty, entry 1 is reserved.

[Table 18](#) shows the entry sequence after a thread is enabled at address A. $[m, n]$ gives the block number in entry 0 and entry 1; x denotes an empty entry.

Table 18. Entry fill sequence

Step	Event	Entries
1	Thread enabled at address A.	$[x, x]$
2	Block containing A written into entry 1; extraction starts at the halfword of A.	$[x, A \gg 5]$
3	Entry 0 empty: the next block is requested when the first request completes.	$[(A \gg 5) + 1, A \gg 5]$
4	Last instruction of entry 1 extracted; entry 1 released.	$[(A \gg 5) + 1, x]$
5	Next block requested into entry 1; extraction continues in entry 0.	$[(A \gg 5) + 1, (A \gg 5) + 2]$

A block write enables only the words from the word containing the request address to word 7. The register file requires its write-data enables one clock phase before the write; the thread computes them in F5 from its token and `f5_icode_resp_miss`, holds them in a phase latch, and applies the word write enables in F6.

2.2.5.4. Instruction Extraction

When both entries are empty, the read pointer is loaded in F5 with the halfword position of the request address within the reserved entry, so that extraction can start within a block. Afterwards the pointer advances with each extracted instruction.

An instruction is available when the entry it is read from is not empty. A 32-bit instruction that starts in halfword 15 requires both entries to be non-empty; the wrap-around read returns both halves, the pointer moves to halfword 1 of the other entry, and the first entry is released.

If the entry being read is empty but its block is written in the same cycle, the instruction is taken directly from the response data (bypass). The upper half of an instruction that crosses into an entry being written is bypassed in the same way. The first instruction of a block is therefore extracted in the cycle the block is written. The bypass does not check the destination entry of the response; with

one request in flight per thread, the only block that can arrive is the one being waited for.

2.2.5.5. RVC Expander

The 32 bits read at the pointer are passed to the RVC expander (`frontend_rvc_expander`), a combinational RV64C decompressor:

- If bits [1:0] are not **11**, the instruction is compressed. It is expanded to its 32-bit equivalent and the pointer advances by 16 bits.
- If bits [1:0] are **11**, the instruction is 32 bits wide. It passes through unchanged and the pointer advances by 32 bits.

All instructions are therefore 32 bits wide after F6. The `rvc` bit of `frontend_core_resp` indicates a compressed instruction; the integer pipeline uses it to advance the PC by 2 instead of 4.

2.2.5.6. Fault and Error Status

Each instruction carries the status of the entries it was read from:

Table 19. Instruction fetch status

Field	Source
<code>pf0, af0</code>	Fault flags of the entry holding the start of the instruction.
<code>pf1, af1</code>	Fault flags of the entry holding the upper half of an instruction that crosses entries; zero otherwise.
<code>bus_error, ecc_error</code>	Error flags of the instruction's entry, or of both entries for an instruction that crosses entries.
<code>replay</code>	Set when the entry's block is not cacheable and <code>f0_core_req.speculative</code> indicates older instructions of the thread in flight. Captured when the instruction is written into the FIFO.

All status fields are forced to zero while the debug program buffer is executed.

2.2.6. Entry Invalidation, Redirect and Thread Start

This section describes the events that discard buffered instructions or restart fetch. [Section 2.2.6.1](#) lists when a buffer entry is released: normally when its last instruction has been extracted, but also on a redirect, a thread disable or entry to debug mode. [Section 2.2.6.2](#) describes a redirect from the integer pipeline: what triggers it, what it clears in each stage, and how long the thread takes to deliver its first target instruction. [Section 2.2.6.3](#) describes how thread enable and disable events are derived from the enable input, how fetch starts at the reset vector, and what state a thread has after reset.

2.2.6.1. Entry Invalidation

An entry is released when its last instruction leaves F6, or when the whole buffer is flushed ([Table 20](#)).

Table 20. Entry release conditions

Condition	Entries released
32-bit instruction at byte offset 28 of the entry is extracted	This entry
Compressed instruction at byte offset 30 is extracted	This entry
32-bit instruction at byte offset 30 is extracted (upper half in the other entry)	This entry
Redirect (<code>f0_core_req_valid</code>) or thread disable	Both entries
Entry to debug mode	Both entries

A response that reports a fault or error marks the reserved entry non-empty even if no data is written, so that the fault is delivered to the integer pipeline with the next instruction read from that entry.

2.2.6.2. Redirect

The integer pipeline redirects a thread by asserting `f0_core_req_valid` with the new PC in `f0_core_req.pc`. Redirects are issued for branch and jump mispredictions, exceptions and returns from trap, replays, inter-processor interrupts, and entry to or exit from debug mode. Table 21 lists the effect in the redirect cycle.

Table 21. Effect of a redirect

Stage	Effect
F0	Fetch PC loaded with the redirect PC. No request in this cycle. Sleep state cleared.
F1	Request waiting in the F1 I-cache request register marked cancelled.
F2–F5	Tokens invalidated.
F6	No block written. Both entries released.
F7	Instruction discarded.
FIFO	Emptied. The thread is not eligible for scheduling in this cycle.

The target block is requested in the following cycle. Its first instruction reaches ID ten cycles after the redirect (Figure 18).

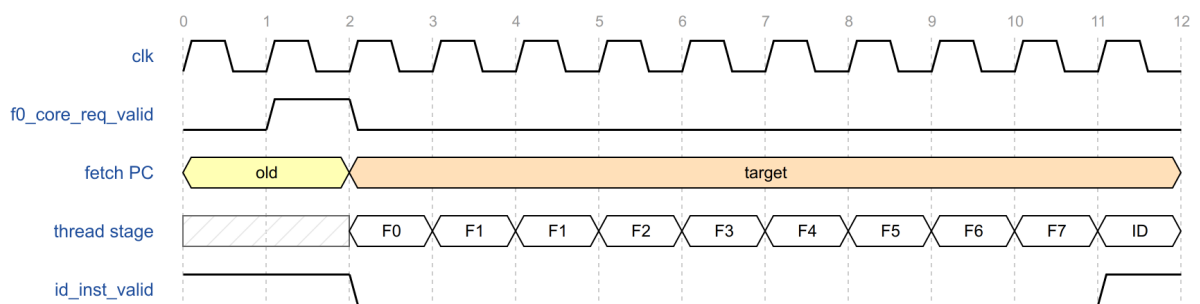


Figure 18. Redirect to first target instruction

2.2.6.3. Thread Enable and Reset Vector

`f0_thread_enabled` is registered on entry to the Front End, and one-cycle enable and disable events are derived from its edges.

- **Enable.** The fetch PC is loaded with `f0_reset_vector` at the end of the enable cycle, and the first request is issued in the next cycle. The first instruction reaches ID eleven cycles after `f0_thread_enabled` rises (Figure 19).
- **Disable.** No further requests are issued, both entries are released and the sleep state is cleared. A fetch already in flight is not cancelled: its response is still written into its reserved entry. The instruction in F7 and the FIFO contents are not cleared. The thread is removed from the scheduler's eligible set one cycle later. Because the scheduler defaults to thread 0, FIFO entries left by a disabled thread 0 can still be presented to ID. The eleven-cycle enable latency applies to a thread whose F7, FIFO and buffer are empty, for example the first enable after reset.

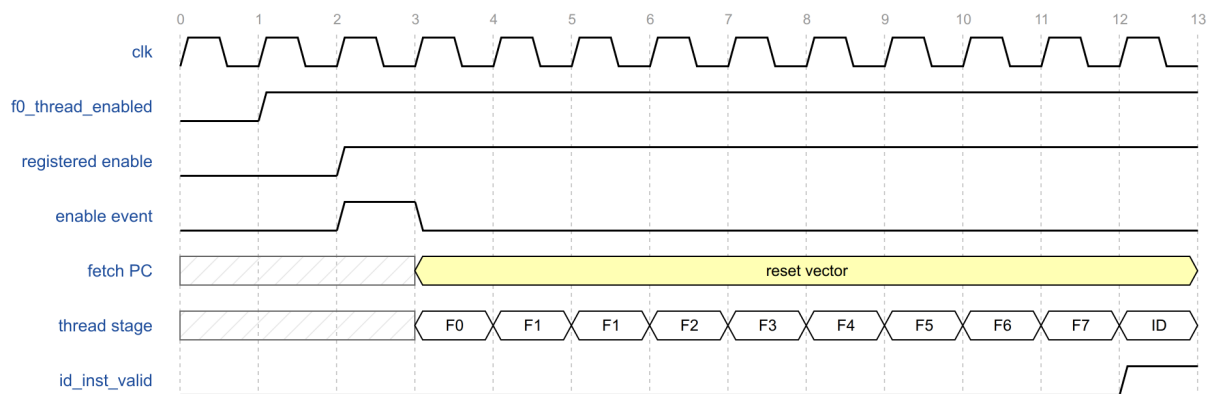


Figure 19. Thread enable to first instruction

After reset, both entries are empty and the fetch PC holds the reset vector.

2.2.7. Decoders

This section describes the integer decoder (`intpipe_decode`) and the VPU decoder (`vpu_decoder`). Both are shared by the two threads and decode in F7, before the instruction is queued. It describes the control words the decoders produce, `minion_control` and `vpu_ctrl_sigs_t`, and how each decoder handles an encoding it does not recognize. It also describes the rule that selects which thread's instruction is decoded in a cycle.

The section also explains how the VPU decode is used once the instruction reaches ID. The Front End does not issue to the VPU itself: the VPU control word goes directly to the VPU. The integer pipeline issues the instruction to the VPU and uses a summary of the control word for its FP and mask scoreboard checks

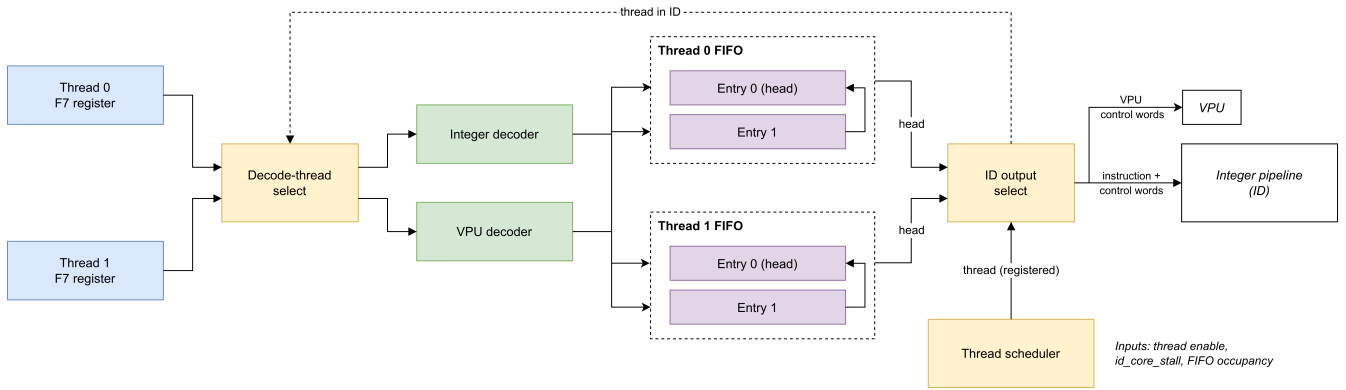


Figure 20. Decoders, instruction FIFOs and thread scheduler

Both decoders are combinational look-up tables over the 32-bit instruction:

- **Integer decoder.** Produces `minion_control` (Table 13). An unlisted encoding is marked not legal, and the integer pipeline raises an illegal-instruction exception.
- **VPU decoder.** Produces `vpu_ctrl_sigs_t` and a flag that identifies VPU instructions. An unlisted encoding is a non-VPU instruction.

A thread is a decode candidate when it has an extracted instruction, its FIFO is not full and it is awake (Section 2.2.9). The decoded thread is selected as follows:

Table 22. Decode thread selection

Candidates	Thread decoded
Thread 0 only	Thread 0
Thread 1 only	Thread 1
Both	Thread not presented in ID in the current cycle
None	Thread not presented in ID in the current cycle

The instruction is written into the FIFO when the selected thread has an extracted instruction and its FIFO is not full; the awake condition affects only the selection.

The Front End does not issue instructions to the VPU. In ID, the VPU control word of the presented instruction (`id_vpu_decoder_sigs`) is sent directly to the VPU, bypassing the integer pipeline. The integer pipeline issues the instruction to the VPU in the same cycle, when the instruction is accepted in ID and decoded as an FP or VPU instruction, together with its thread and that thread's fcsr rounding mode. The integer pipeline also receives a summary of the VPU control word (`id_vpu_core_ctrl`), with the VPU and mask register reads and writes of the instruction, for its FP and mask scoreboard checks.

2.2.8. Instruction FIFO

This section describes the two-entry instruction FIFO that each thread has (uses) between F7 and ID. The FIFO decouples fetch from issue. While ID is stalled or serving the other thread, the thread keeps extracting and decoding, and the FIFO holds the decoded instructions with their fetch status. The section covers what an entry holds and the pointer scheme that tracks occupancy. It explains why the storage is arranged for timing, as a fixed head register and a holding register, and gives the

update rules. It also covers when the VPU control word is valid and how a redirect empties the FIFO.

The FIFO is written when its thread is selected for decode and read when the integer pipeline accepts the thread's instruction (`id_inst_valid`, `id_inst_ready` and `id_inst_thread_id`). It uses 2-bit read and write pointers; it is full when the pointers differ only in the wrap bit and empty when they are equal.

The storage is organized for timing rather than as an indexed array. Entry 0 is always the head, whose content drives the ID outputs; entry 1 is a holding register. The ID multiplexer therefore selects between two fixed registers, one per thread.

Table 23. Instruction FIFO update

Occupancy	Event	Update
Empty	Write	Entry 0 $\leftarrow$ new instruction
One	Write and read	Entry 0 $\leftarrow$ new instruction
One	Write	Entry 1 $\leftarrow$ new instruction
One	Read	None; FIFO becomes empty
Full	Read	Entry 0 $\leftarrow$ entry 1

The VPU control word of an entry is written only for VPU instructions. For other instructions it retains the value of an earlier VPU instruction, so `id_vpu_decoder_sigs` and `id_vpu_core_ctrl` are valid only for VPU instructions.

A redirect empties the thread's FIFO.

2.2.9. Thread Scheduler

The integer pipeline accepts one instruction per cycle from either thread. This section describes how the thread scheduler (`frontend_thread_sched`) chooses which thread's FIFO head is presented to ID in the next cycle. It defines when a thread is awake (enabled and not stalled by the integer pipeline) and when it is ready (it has an instruction and is not being redirected). It gives the round-robin rule between eligible threads, including the default choice when neither thread is eligible. It then describes the handshake with ID: what drives the ID outputs, why the valid signal is not qualified with the stall, and what happens to an instruction that ID does not accept.

Awake

The thread is enabled and `id_core_stall` is deasserted. Awake is registered, so a stall affects scheduling one cycle after it is asserted.

Ready

The thread's FIFO is not empty or is being written in the current cycle, and the thread is not being redirected. Including the write allows an instruction to pass from F7 to ID in consecutive cycles.

A thread that is awake and ready is eligible. The scheduler selects between eligible threads in round-robin order:

Table 24. Thread selection

Thread 0 eligible	Thread 1 eligible	Selected
Yes	No	Thread 0
No	Yes	Thread 1
Yes	Yes	Thread not selected previously
No	No	Thread 0

The previous selection is updated only when at least one thread is eligible. The selection is registered and drives `id_inst_thread_id` in the next cycle.

The head of the selected thread's FIFO drives `id_inst_data`, `id_intpipe_ctrl`, `id_vpu_decoder_sigs` and `id_vpu_core_ctrl`. `id_inst_valid` is asserted when that FIFO is not empty, and the integer pipeline accepts the instruction with `id_inst_ready`. `id_inst_valid` is not qualified with `id_core_stall`; the integer pipeline applies the stall. The thread in ID is reselected every cycle, so an instruction that is not accepted can be replaced by the other thread's instruction in the next cycle; it remains at the head of its FIFO.

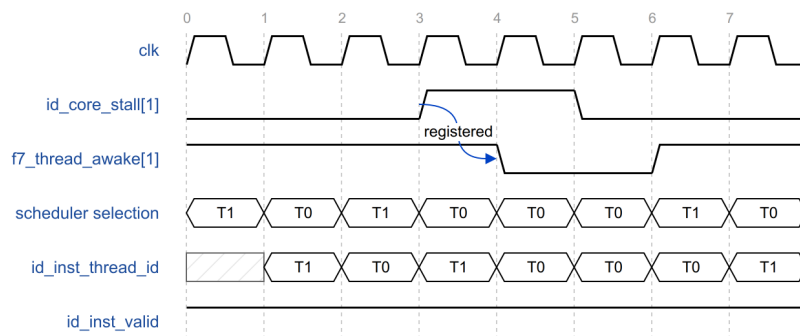


Figure 21. Round-robin issue with a stall

In Figure 21 both threads are eligible and alternate. `id_core_stall` of thread 1 is asserted in cycles 3 and 4, and thread 1 is not awake in cycles 4 and 5. Thread 0 is selected in cycles 3 to 5 and presented in ID in cycles 4 to 6; the selection alternates again from cycle 6.

2.2.10. Halt and Debug Program Buffer

This section describes the Front End's two debug functions. Section 2.2.10.1 covers halt. The integer pipeline takes a halt only on a valid instruction in ID, so while a halt is requested the thread buffer generates a placeholder instruction instead of fetching, and passes it through the normal pipeline to ID.

Section 2.2.10.2 covers the debug program buffer. In debug mode, entry 0 of the halted thread's buffer holds the instructions written by the debugger. The section describes how the buffer is prepared with **EBREAK** on entry to debug mode, how the debugger writes words into it and starts execution, and how a resume redirect ends program-buffer mode. It also shows which write data goes to which word, and why the last word always holds **EBREAK**.

2.2.10.1. Halt

The integer pipeline takes a halt on a valid instruction in ID. While `halt` is asserted for a thread, the thread buffer generates a placeholder instruction instead of fetching:

1. The thread follows the normal request conditions but does not bid for the request port. The request is treated as accepted internally, and its token enters the tracker.
2. The token passes through F1 to F6.
3. In F6 an entry write is forced without a response. The data is the last registered response data; fault and error flags are cleared, and the instruction is treated as 32 bits wide.
4. The placeholder passes through decode and the FIFO to ID, where the integer pipeline takes the halt.

The content of the placeholder is undefined. A pending halt also clears the sleep state.

2.2.10.2. Debug Program Buffer

In debug mode, entry 0 of the halted thread's buffer is used as the program buffer: the debugger writes instructions into it and the thread executes them.

Table 25. Debug program-buffer sequence

Step	Operation
Entry	A redirect with <code>debug_info.halt</code> arrives; the thread stops fetching in the same cycle. In the next cycle both entries are released, the extracted instruction is discarded, and words 0 to 4 of entry 0 are written with <code>EBREAK</code> .
Write	The debugger writes the program buffer through <code>debug_ffb_wdata</code> , <code>debug_ffb_en</code> and <code>debug_ffb_thread_sel</code> (Table 26). The data and enables come together but the data is written into the entry in the cycle after the enables.
Execute	<code>debug_ffb_exec</code> resets the read pointer to halfword 0 of entry 0. In the next cycle entry 0 is marked non-empty and entry 1 empty, and instructions are extracted from word 0.
Exit	A redirect with <code>debug_info.resume</code> ends program-buffer mode; fetch restarts at the redirect PC.

Table 26. Program-buffer word mapping

Word	Write enable	Data
0	<code>debug_ffb_en[0]</code>	<code>debug_ffb_wdata[31:0]</code>
1	<code>debug_ffb_en[1]</code>	<code>debug_ffb_wdata[63:32]</code>
2	<code>debug_ffb_en[2]</code>	<code>debug_ffb_wdata[31:0]</code>
3	<code>debug_ffb_en[3]</code>	<code>debug_ffb_wdata[31:0]</code>
4	—	<code>EBREAK</code>

Word 4 cannot be written by the debugger and always holds `EBREAK`, so execution returns to debug mode at the end of the program buffer. While the program buffer is executed, all fault, error and

replay status is forced to zero. A debug reset while the thread is halted rewrites words 0 to 4 with **EBREAK**.

2.3. Integer Pipeline

2.3.1. Overview

Figure: Integer pipeline block diagram.

2.3.2. Pipeline Stages

Figure: Integer pipeline stages (ID, EX, TAG, MEM, WB).

2.3.3. Interfaces

Table: Integer pipeline interface signals.

2.3.4. Register File and Operand Read

2.3.5. Immediate Generation and ALU

2.3.6. Multiply/Divide Unit

2.3.7. Scoreboards and Hazards

2.3.7.1. Integer Scoreboard

2.3.7.2. FP Scoreboard

2.3.7.3. Mask Scoreboard

2.3.8. Hart Interleaving and Stalls

2.3.9. Branches, Redirects and Replay

2.3.10. Interface to the Data Cache

2.3.11. Interface to the VPU

2.3.12. CSRs and Privilege Modes

Table: CSR groups.

2.3.13. Exceptions and Interrupts

2.3.14. Debug and Breakpoints

2.3.15. Performance Counters

2.3.16. Verification Notes

2.4. Data Cache

2.4.1. Overview

Figure: Data cache block diagram.

2.4.2. Organization and Operating Modes

Table: Data cache operating modes.

2.4.3. Interfaces

Table: Data cache interface signals.

2.4.4. Request Pipeline

Figure: Data cache request pipeline.

2.4.5. Address Translation and PMA Checks

2.4.6. Tag, Data and LRU Arrays

2.4.7. Miss Handling

2.4.8. Replay Queue

2.4.9. Writeback and Store Merge

2.4.10. Atomic Operations

2.4.11. Cache-Management Operations

2.4.12. Scratchpad Control

2.4.13. Tensor Load and Reduce

2.4.14. Interface to the Neighborhood

2.4.15. Errors

2.4.16. Performance

2.4.17. Verification Notes

2.5. Vector Processing Unit

2.5.1. Overview

Figure: VPU block diagram.

2.5.2. Interfaces

Table: VPU interface signals.

2.5.3. Control and Micro-Instruction Decode

2.5.4. Register Files

2.5.5. Lanes

Figure: VPU lane.

2.5.6. Mask Unit

2.5.7. Bypass Network

2.5.8. Shift and Swizzle Unit

2.5.9. TxFMA Unit

2.5.10. Integer Multiply-Accumulate Unit

2.5.11. Transcendental Unit

2.5.12. Tensor Operations

2.5.12.1. Tensor FMA

2.5.12.2. Tensor Quantization

2.5.12.3. Tensor Reduce

2.5.13. Interfaces to the Integer Pipeline and Data Cache

2.5.14. Performance

2.5.15. Verification Notes

3. Neighborhood

3.1. Neighborhood Overview

3.1.1. Partitioning

Figure: Neighborhood block diagram.

3.1.2. Shared Resources and Core Connections

3.1.3. Voltage-Domain Crossings

A Neighborhood is divided into two clock and voltage domains:

- **Low-voltage (LV) region:** the eight ET-Minion cores and most of the Neighborhood channel, running on the Neighborhood clock.
- **High-voltage (HV) region:** the logic that interfaces to the Shire Channel, running on the Shire clock.

Blocks were assigned to the two regions for performance and power: the cores and the channel logic run at the lower voltage to save power, while the logic and the memory cells that need speed run at the higher one. Two properties of an SRAM cell decide where its memory has to go, and both improve with voltage:

- **Access time.** A cell driven from a higher supply switches its bit lines faster, so the array is read and written in less time.
- **Noise margin.** Noise margin is the amount of disturbance, from coupling between neighbouring lines, supply droop or process variation, that a cell can absorb before the value it holds or the value being read out flips. It shrinks as the supply falls, so reads and writes become less reliable. A memory macro is therefore characterized with a minimum operating voltage, and carries trim settings, read assist, write assist and write-pulse width, that buy margin back at low voltage at the cost of speed.

Memory cells were kept at the higher voltage for both reasons, which is also why the cells of the crossing FIFOs themselves are always placed in the high-voltage region, whichever direction the interface goes. Margin is not the same as immunity, so the data path from the memory is protected as well: the line returned from the L1 instruction cache data RAM is covered by ECC, the error is reported to the requesting core with the response and it is logged in the Neighborhood ESRs ([Section 3.3.6](#)).

The two regions are connected by a semi-synchronous interface, and every path between them passes through voltage-crossing FIFOs, whose crossing elements contain the level shifters. The paths that cross are the memory request and response paths, the fast local barrier path, the debug and ESR accesses, and the SRAM interface of the L1 instruction cache, whose data RAM is in the high-voltage region of the Shire Channel. Each crossing costs latency, which is why these paths are longer than their logic alone suggests.

3.1.4. External Interfaces

Table: Neighborhood interface groups.

3.2. Neighborhood Instruction Cache

A Neighborhood groups eight ET-Minion cores. Each core has its own private L1 data cache, but none of them has a private instruction cache. Instead, the eight cores share one instruction cache, a 32 KB L1 I-Cache that the Neighborhood adds as a shared resource. The cores of a Neighborhood normally execute the same code, so a shared cache holds one copy of it instead of eight, and that saves both area and power.

The shared instruction cache has two levels of access because of the voltage region its memory has to be built in. SRAM cells are faster and hold their contents more reliably at a higher supply voltage, so the cache's data RAM is placed in the high-voltage region, while the Neighborhood itself operates at low voltage ([Section 3.1.3](#)).

The cache's data RAM is therefore placed outside the Neighborhood, in the Shire Channel, and every access to it passes through the voltage-domain crossing, which lengthens the access time. Serving every core from that cache alone would also leave all 16 threads of the Neighborhood, eight cores with two threads each, competing for its single request port. Access to that shared port is handled by LRU arbitration, so a thread would wait behind up to 15 others for its turn, even when the line it needs is already present.

To overcome both, the long read latency of its data RAM in another voltage region and the wait for a turn among 16 bidding threads, a micro level was added in front of that shared cache: two small first-level caches, called L0 micro-caches, inside the Neighborhood, next to the cores. Their data arrays are built in the Neighborhood channel, so a hit is served without crossing into the Shire Channel, and each micro-cache serves half the cores, so eight threads share a port instead of 16.

The shared instruction cache of the Neighborhood is thus split into the following hierarchy:

- **L0 micro-caches:** 16 fully associative lines of 64 bytes each, with their own instruction TLB, PMA check, LRU victim selection and page-table walker port ([Section 3.6](#)), and a five-stage lookup. There are two of them, called the north and the south micro-cache, serving Minions 0 to 3 and Minions 4 to 7 respectively, that is eight threads each. Both are identical except for one function, the code prefetcher, enabled in the north micro-cache only, for fetching cache lines into the L1 instruction cache.
- **L1 instruction cache:** 32 KB, 128 sets x 4 ways, 64-byte lines, shared by both micro-caches and accessed only when one of them misses. Its tags, LRU and control are inside the Neighborhood; its data RAM is the one in the Shire Channel.

[Figure 22](#) shows the structure. [Section 3.2.1](#) describes how the two cache levels are packaged in the RTL and what blocks they connect to. The micro-caches and the L1 instruction cache are described in detail in [Section 3.4](#) and [Section 3.5](#), and the path from the core's Front End to the cache, with its arbitration and priority updates, in [Section 3.3](#).

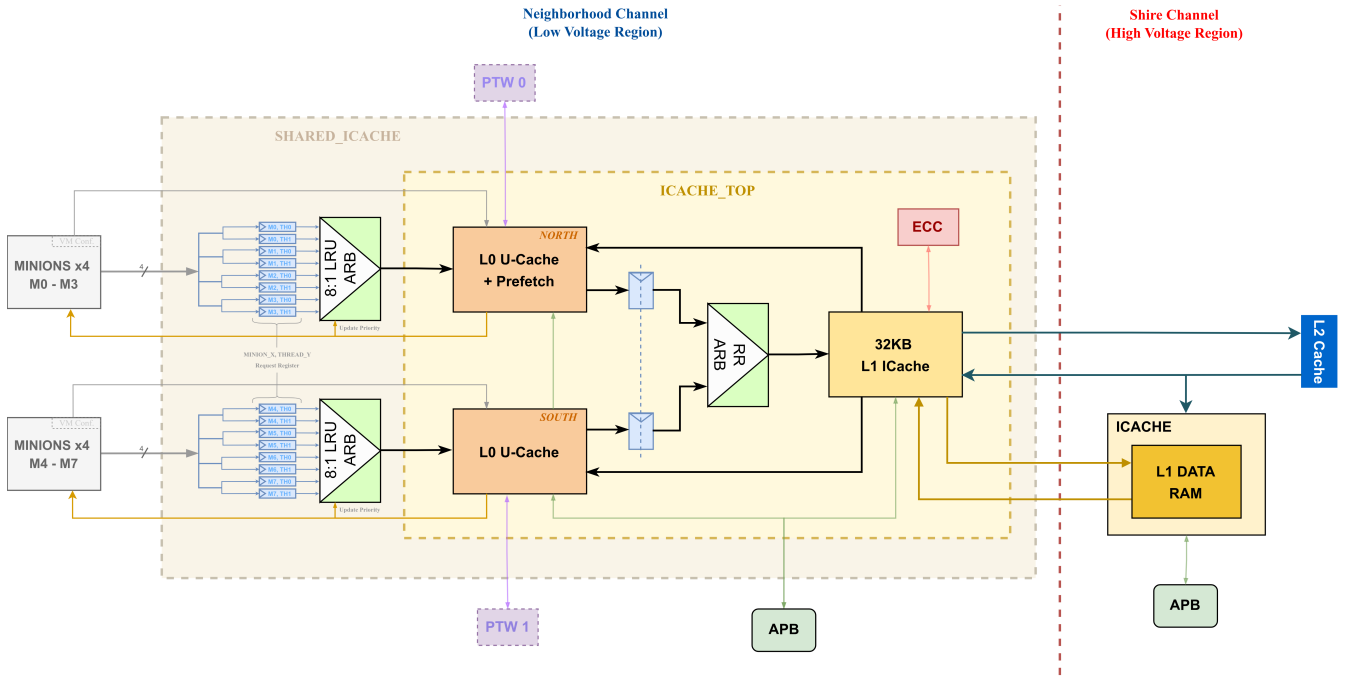


Figure 22. Neighborhood instruction cache

3.2.1. Structure and Connections

The cache is implemented in two modules:

- **icache_top** contains both L0 micro-caches and the L1 instruction cache, everything except the L1 data RAM, that is the micro-cache arrays and pipelines, the instruction TLBs, the PMA checks, the L1 tag, LRU and data-array control, and the round-robin arbiter through which the two micro-caches reach the L1.
- **neigh_shared_icache** wraps **icache_top** and adds the Neighborhood-side logic of the core interface, the logic that receives the fetch request of each of the eight cores, holds it per thread and presents one of them to a micro-cache: one request register per thread, the 8:1 LRU arbiter that selects among them, and the per-thread miss counters and bid mask that prevent livelock. These are described in [Section 3.3](#).

The top-level **neigh_shared_icache** block connects to five things outside itself:

Interface	Purpose
Cores	Fetch requests and address-translation configuration, one port per core. Responses and the fill-done indication, one port per L0 micro-cache, each driven to the four cores that micro-cache serves.
Page-table walkers	Translation requests from the instruction TLB of each micro-cache, one walker per micro-cache.
Shire Cache	Line requests on an L1 miss, and their responses, over the Neighborhood request and response path.
L1 data RAM	SRAM read and write requests to the data RAM in the Shire Channel, and the read data returned from it.
ESR and APB	Configuration (prefetch, bypass, protection and page size), error logging, and debug access to the arrays.

The signals of each interface are listed with the block that uses them: the core-side signals in [Section 3.3.2](#), and the rest in the detailed sections of the two cache levels.

3.3. Front End – I-Cache Interface

This section describes the fetch protocol between the ET-Minion cores and the instruction cache, from the Neighborhood side. The core side is described in [Section 2.2.4](#).

3.3.1. Overview

The eight cores of a Neighborhood are served by two L0 micro-caches, each shared by four cores, that is by eight threads, two threads per core. Both L0 micro-caches are backed by the L1 instruction cache of the Neighborhood. [Figure 23](#) shows the whole fetch path and names the signals of the interface: the two thread buffers and the 2:1 LRU arbiter inside each core, the per-thread request registers and the 8:1 LRU arbiter of each micro-cache, and the round-robin arbiter through which both micro-caches reach the L1 instruction cache.

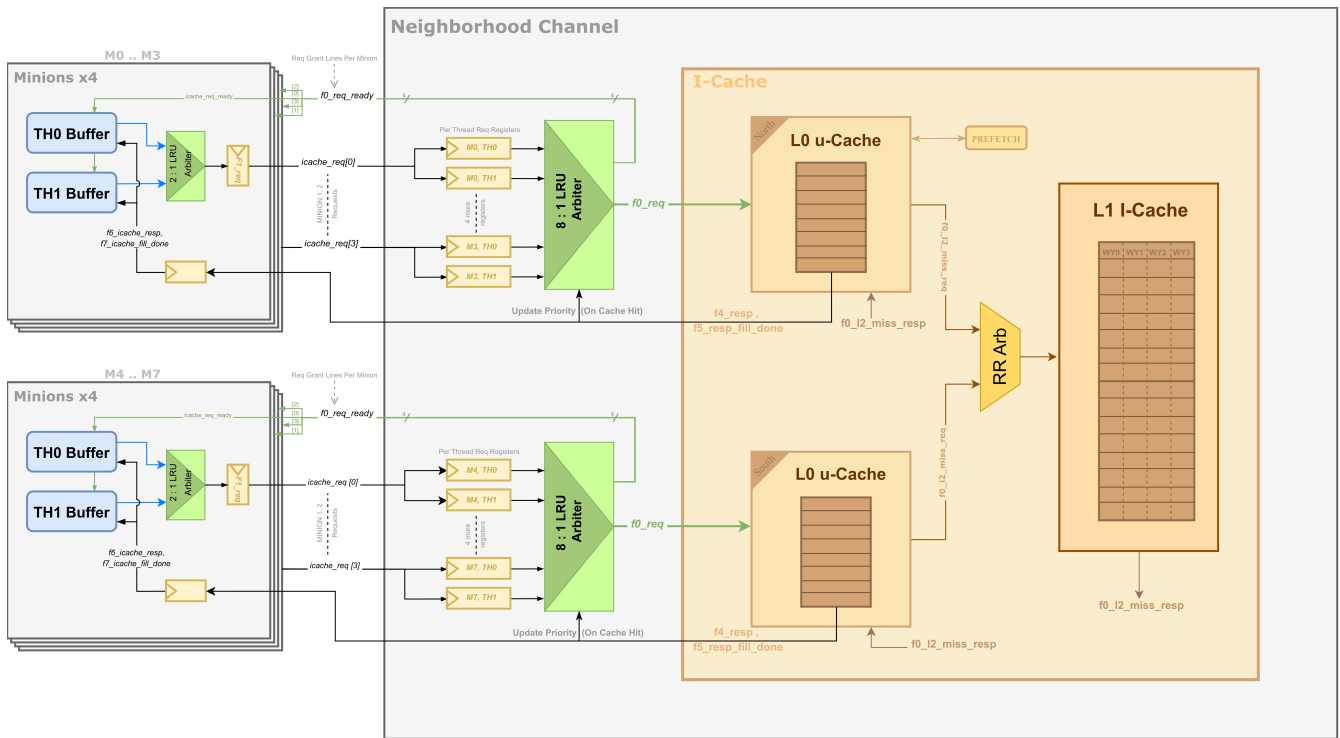


Figure 23. Front End – I-cache interface

A fetch request, that is a request from the Front End of a core for the instruction block at a thread's fetch address, travels as follows:

1. In the core, the 2:1 LRU arbiter selects one of the two threads, and the selected request is flopped into the Front End's F1 request register, which holds it until the Neighborhood accepts it. Each core therefore presents one request at a time.
2. In the Neighborhood, the request is steered into the request register of the thread that made it, using the thread identifier it carries. One register exists per thread, so eight per L0 micro-cache, which lets the threads of all four cores arbitrate on equal terms.
3. The 8:1 LRU arbiter selects one of the eight captured thread requests and presents it to the L0 micro-cache. The grant is returned to the core in the same cycle, as `icache_req_ready`, and it

frees the core's F1 request register: the core's 2:1 arbiter runs in that same cycle and loads the other thread's request, so the core presents it on the next cycle with no idle cycle in between.

4. The L0 micro-cache looks the request up in five stages: arbitration, instruction-TLB translation, tag lookup and PMA check, data-array read, and response.
5. One response is returned a fixed four cycles after the grant, for a hit and for a miss alike, and it reaches the core in its F5 stage. It is shared by the eight threads the micro-cache serves and carries no thread identifier; on a hit it carries the instruction block. How a core recognizes its own response is described in [Section 2.2.4.2](#).
6. On a miss, which can be a TLB miss or a line miss, the L0 micro-cache requests a translation from its page-table walker, or requests the line from the L1 instruction cache through a round-robin arbiter that the two micro-caches share. When the fill completes, the micro-cache pulses fill done to all the threads it serves, and the sleeping ones wake and retry.

3.3.2. Interface Signals

[Table 27](#) lists the signals between a single core and the Neighborhood. Directions are given from the Neighborhood side.

Table 27. Front End – I-cache interface signals

Signal	Dir	Width / type	Description
Fetch request			
<code>icache_req_valid</code>	in	1	The core presents a fetch request.
<code>icache_req</code>	in	<code>frontend_icache_req</code>	Requesting thread, its virtual-memory status and the fetch address.
<code>icache_req_ready</code>	out	1	The 8:1 arbiter has granted the request. Returned in the grant cycle, it frees the core's F1 request register, so the core can load its other thread's request in the same cycle and present it on the next.
Fetch response			
<code>icache_resp_valid</code>	out	1	A response is returned in this cycle.
<code>icache_resp_miss</code>	out	1	The response is a miss and carries no instruction data.
<code>icache_resp</code>	out	<code>icache_frontend_resp</code>	32-byte block with its fault, error and cacheable status.
<code>icache_fill_done</code>	out	1	A line or translation fill has completed in the L0 micro-cache.
Address translation and maintenance			
<code>satp_info</code> , <code>matp_info</code>	in	2 × <code>minion_satp_info</code>	Supervisor and machine address-translation registers of the core, used by the instruction TLB.

Signal	Dir	Width / type	Description
<code>tlb_invalidate</code>	in	1	Invalidate the instruction-TLB entries of this core.
<code>icache_flush_data</code>	in	1	Invalidate the data stored in the L0 micro-cache.

The fetch response ports are the exception to that one-core view. The Neighborhood drives one set of them per L0 micro-cache, common to the four cores it serves, so `icache_resp_valid`, `icache_resp_miss`, `icache_resp` and `icache_fill_done` reach all four at the same time. Being common, the response carries no thread identifier and nothing that says which core it belongs to. Each core works that out on its own: it tracks its outstanding request down the pipeline, so it knows in which cycle its response is due and which of its two threads made it, and it ignores the responses of the other three cores ([Section 2.2.4.2](#)).

3.3.3. Request Arbitration

This section describes how the Neighborhood decides which thread's request reaches the L0 micro-cache in a given cycle, and when the arbiter updates the priority that decision is based on. Lets start with the registers that hold a thread's request on its way in.

A fetch request is flopped twice before it enters the L0 micro-cache.

First flop, in the core. Two threads share one fetch port of the core, so the 2:1 LRU arbiter selects one of them and its request is held in the Front End's F1 request register. That register keeps the request until the Neighborhood (the L0 micro-cache in this case) accepts it, so a core presents one request at a time, carrying the identifier of the thread that made it.

Second flop, in the Neighborhood. Each micro-cache serves four cores, so it sees at most four requests in a cycle, and it holds one request register per thread, eight in all. Each request is written into the register of the thread its identifier names. At most four of the eight registers are valid at a time, because a core frees its F1 register only when its request is accepted. Eight registers exist nevertheless so that every thread bids for itself, which is what makes the arbitration fair between the two threads of a core as well as between cores.

The eight registers bid through the 8:1 LRU arbiter (`arb_lru_delayed_data`). A bid is the register's valid qualified with the thread's bid mask. The mask exists to avoid livelock and is described in [Section 3.3.8](#); until then, take it that masking suppresses a thread's valid at the arbiter, so the thread takes no part in the arbitration while its request stays in its register.

The arbiter is stalled while the micro-cache cannot accept a request, which the micro-cache reports with `f0_req_ready`. Its pipeline never stalls, so this signal is tied high: a request is accepted in every cycle in which at least one thread bids, and the rest of this section takes that for granted.

3.3.3.1. Priority Matrix

The 8:1 LRU arbiter holds its priority in an aging matrix: one bit for every pair of the eight threads it serves, recording which of the two was served more recently. A thread wins when it has priority over every other thread bidding in that cycle, which makes the grant one-hot. Two threads can

never hold the same priority, because every pair is ordered by exactly one bit. Granting a thread clears all of its bits, that is it is placed behind every other thread, and the threads that have waited longest move up.

The grant is combinational. In the same cycle it is returned to the core as `icache_req_ready`, the thread's request register is freed, so it reads invalid in the next cycle, and the core may load the request of its other thread into the F1 register. A request is therefore accepted at the earliest in its second cycle in the Front End's F1 stage, because that register has to hold it for one more cycle while it is flopped into the Neighborhood's per-thread register.

3.3.3.2. Deferred Priority Update

Every accepted request gets exactly one response, and it comes back a fixed four cycles after the grant, whether it hits or misses (Section 3.3.4). The arbiter waits for it: priority is not updated when the grant is issued, but when that response arrives.

Waiting means the arbiter has to remember whose response is due, because the response is shared and carries no thread identifier. The grant vector therefore travels down with the request in its own registers, `f1_grant` to `f4_grant`, so that in F4 it can be told which thread the response belongs to (Table 28).

Table 28. Grant pipeline in the Neighborhood

Stage	Register	Content
F1	<code>f1_grant</code>	One-hot grant of the arbiter, loaded in the cycle the L0 micro-cache accepts the request
F2	<code>f2_grant</code>	Same vector, one cycle later
F3	<code>f3_grant</code>	Same vector, one cycle later
F4	<code>f4_grant</code>	Thread whose response is returned in this cycle; selects the priority bits and the miss counter to update

The update is enabled by `f4_update_prio`, asserted when a response is valid in F4 and that response is a hit, or reports a page fault or an access fault:

- **Hit.** Priority is updated and the thread moves behind the others, which gives the waiting threads their turn.
- **Miss.** Priority is left unchanged, so the thread keeps its position while it sleeps and is served early when it retries after the fill (Figure 25).
- **Page fault or access fault.** Priority is updated, because the thread does not retry the same fetch: the core records the fault with the instruction and takes the trap (Section 3.3.6).

Bus and ECC errors do not appear in this condition. Both are reported only with a hit, and a hit already updates the priority.

3.3.4. Request and Response Timing

The L0 micro-cache looks a request up in five stages and drives fill done one stage later (Table 29).

Table 29. L0 micro-cache pipeline stages

Stage	Operation
F0	Arbitration; the granted request enters the pipeline.
F1	Instruction-TLB lookup, which translates the virtual address.
F2	Tag lookup, PMA check and LRU update. A miss is detected here, and accepted if the miss handler is idle.
F3	Data array read. On an accepted miss, the request to the L1 instruction cache is driven in this cycle.
F4	Response to the cores (<code>f4_resp_valid</code> , <code>f4_resp_miss</code> , <code>f4_resp</code>).
F5	Fill-done pulse to the cores (<code>f5_resp_fill_done</code>), one stage after the response. Not part of the lookup: it reports that a line or a translation has been written.

These stages are the micro-cache’s own, and they are numbered independently of the Front End’s F0 to F7: the F1 request register of the core and the F1 stage of the micro-cache are different things.

Every accepted request receives exactly one response from the last stage, for a hit and for a miss alike, so the response arrives a fixed four cycles after the grant. The response is broadcast to the four cores of the L0 micro-cache and carries no thread identifier; each core identifies its own response from the position of its request in time ([Section 2.2.4.2](#)).

The fill-done signal is driven one stage later than the response, from the stage after F4.

In the waveforms that follow, a thread is written M0, TH0, meaning thread 0 of Minion 0. [Figure 24](#) shows a request that hits: the core presents it, the 8:1 LRU arbiter grants it in cycle 1, and the response is returned in cycle 5, four cycles later. The grant frees both registers that were holding the request, the core’s F1 request register and the Neighborhood’s per-thread register, which is why both valids fall in the next cycle. `f4_grant` still names M0, TH0, the thread that was granted four cycles earlier, which is how F4 knows whose priority to update: the response is a hit, so `f4_update_prio` is asserted and M0, TH0 moves behind the other threads in the LRU order.

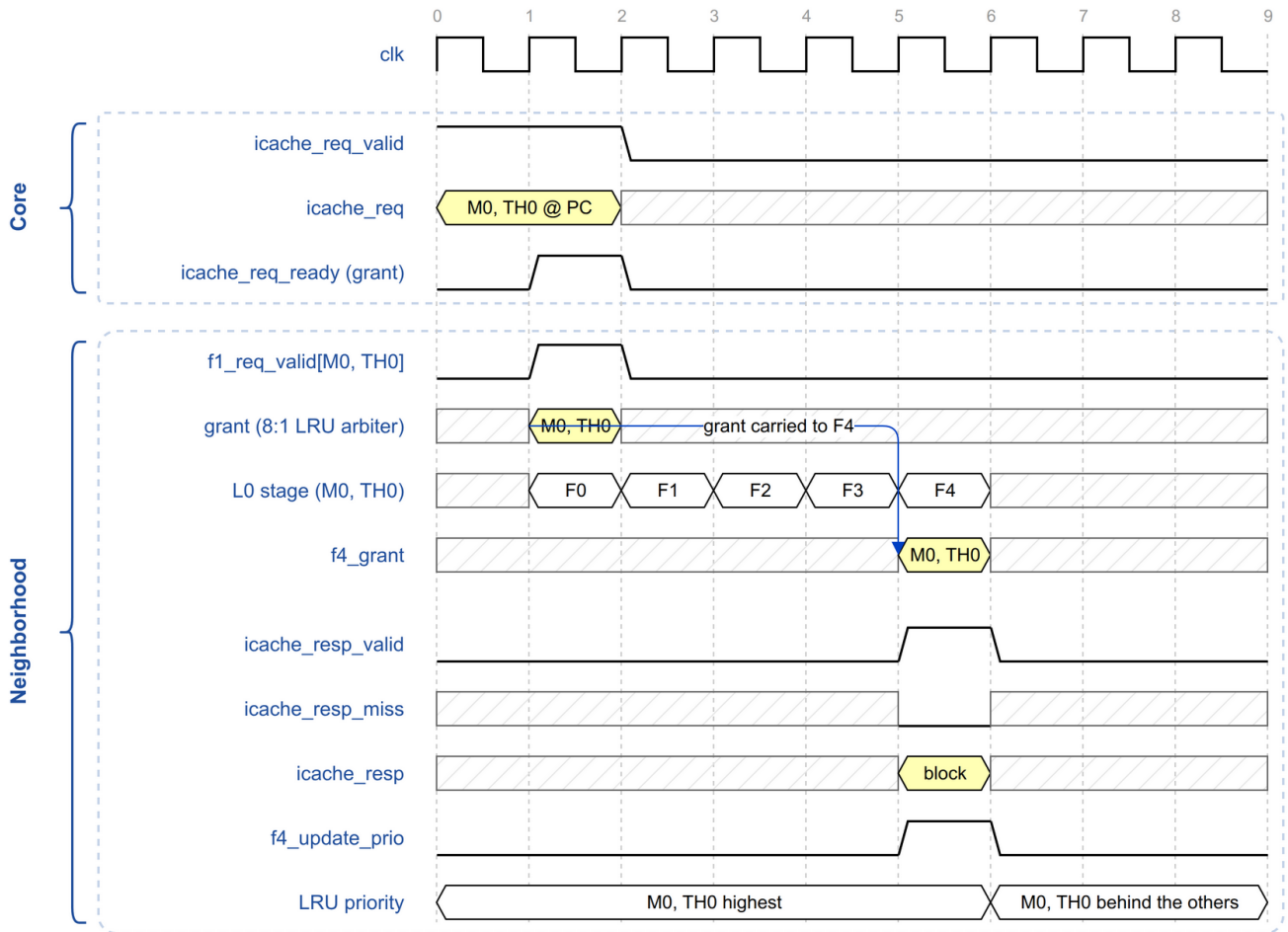


Figure 24. Request and hit response at the L0 micro-cache

3.3.5. Miss, Sleep and Fill-Done Protocol

A request misses in the L0 micro-cache either because the instruction TLB has no translation for it, or because the line is not in the micro-cache:

- **TLB miss.** A request is sent to the shared page-table walker (Section 3.6), and the lookup continues down the pipeline, so the requesting thread receives its miss response with the normal timing.
- **Tag miss.** A request for the line is sent to the L1 instruction cache. The two L0 micro-caches share one request port to the L1 and are selected by a round-robin arbiter, each with a holding register for its request.

In both cases the core that made the request receives a miss response and puts the requesting thread to sleep.

When the fill completes, that is when the translation has been written into the TLB or the line into the micro-cache data array, the L0 micro-cache pulses `icache_fill_done`. The pulse is common to all eight threads it serves and carries no address, so every sleeping thread wakes and retries. A thread whose line is still absent misses again and sleeps until the next fill.

Fill done is deliberately driven one stage later than the response. A request that looks the cache up while the fill is being written still misses, and its miss response is the last one that this fill can cause. Sending fill done after that response guarantees that every thread has entered the sleep state

before its wake-up arrives, so no wake-up is lost.

Each L0 micro-cache has one miss handler, which holds one miss at a time; it is described in [Section 3.3.8.1](#). [Figure 25](#) follows one thread through a miss and its retry. The lookup misses in F2, the handler leaves **Ready**, and the request to the L1 instruction cache is driven in F3. The response in F4 is a miss, so the thread sleeps and its priority is left unchanged; the grant has already freed both the core's F1 request register and the Neighborhood's per-thread register. When the line arrives the handler returns to **Ready** and fill done is pulsed; the thread wakes, requests again and hits, and only that second response updates the priority.

The break in the waveform stands for the access time of the L1 instruction cache, and of the Shire Cache behind it if the L1 misses as well. That time is not fixed and is far longer than the rest of the sequence. The handler stays in **Fill_Wait** for all of it, so every other lookup that misses meanwhile is turned away, and the core leaves the thread asleep and issues nothing for it. When the line arrives the handler returns to **Ready** and fill done is pulsed; the thread wakes, requests again and hits, and only that second response updates the priority.

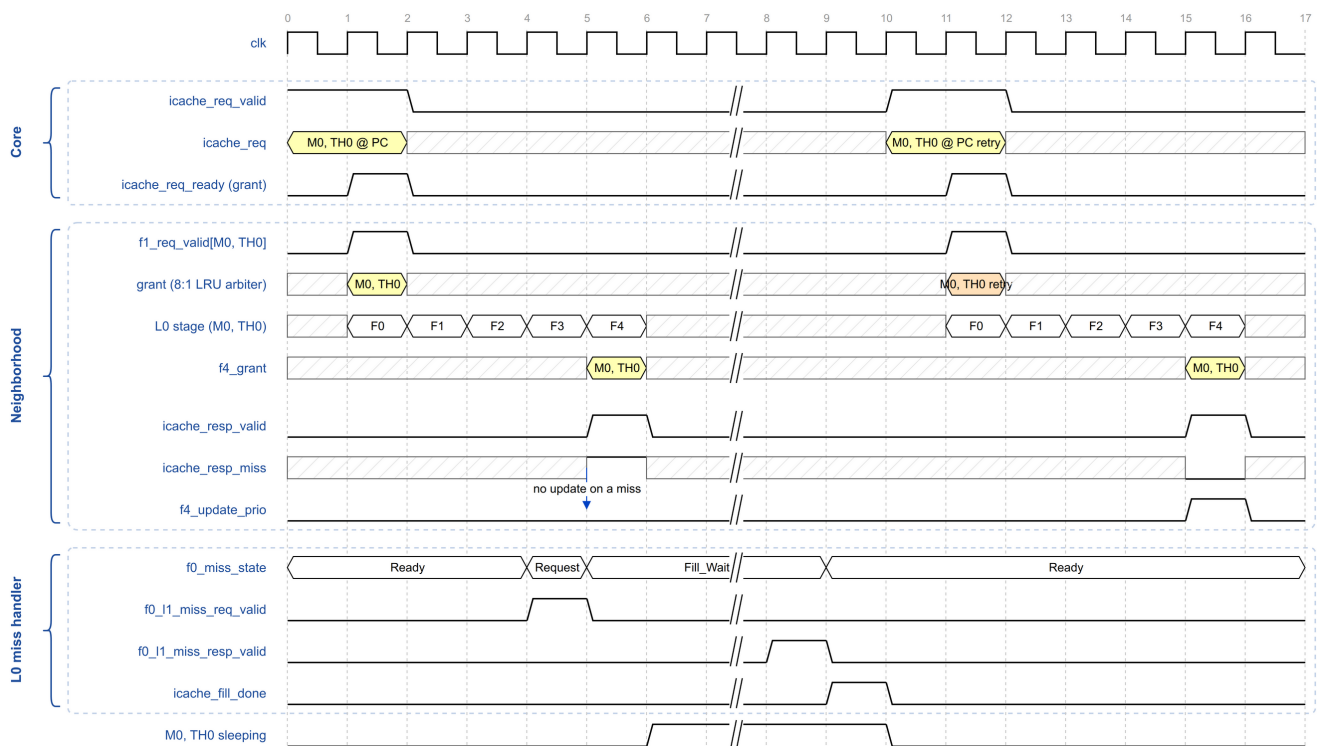


Figure 25. Miss, sleep, fill done and retry

3.3.6. Fault and Error Reporting

The response carries the fault and error status of the fetch together with the data:

Field	Meaning
page_fault	The instruction TLB could not translate the address, or the translation forbids the access. Reported whether the lookup hit or missed.
access_fault	The PMA check rejected the physical address. Reported whether the lookup hit or missed.
cacheable	The block comes from a cacheable region, as reported by the PMA check. Valid on a hit.

Field	Meaning
<code>ecc_err</code>	The line was filled with an ECC error. The micro-cache has no ECC of its own: it stores the error flag that arrived with the fill next to the line and returns it on every hit on that line, until the line is replaced. Only the error classes enabled in the error-log control ESR set the flag. Valid on a hit.
<code>bus_err</code>	The line was returned with an error by the memory hierarchy. Valid on a hit.

A response that reports a fault is delivered like any other response, and the core marks the faulting instruction so that the integer pipeline raises the exception ([Section 2.2.5.6](#)). Errors detected in the L1 instruction cache data RAM are also logged in the Neighborhood ESRs, which hold a sticky single-bit and double-bit error flag and, for the access that was logged, its way, set and physical address and the ECC blocks of the line in which the error was found.

3.3.7. Complete Path for Four Cores

[Figure 26](#) puts the pieces together: four cores requesting in the same cycle, the two flops each request passes through, the arbitration between the threads, and the five responses, one for each of the five threads that were granted. In the `LRU_order` lane alone the threads are written as digits, numbered $2 \times \text{Minion} + \text{thread}$, because eight names do not fit in one cycle; the lane lists them highest priority first.

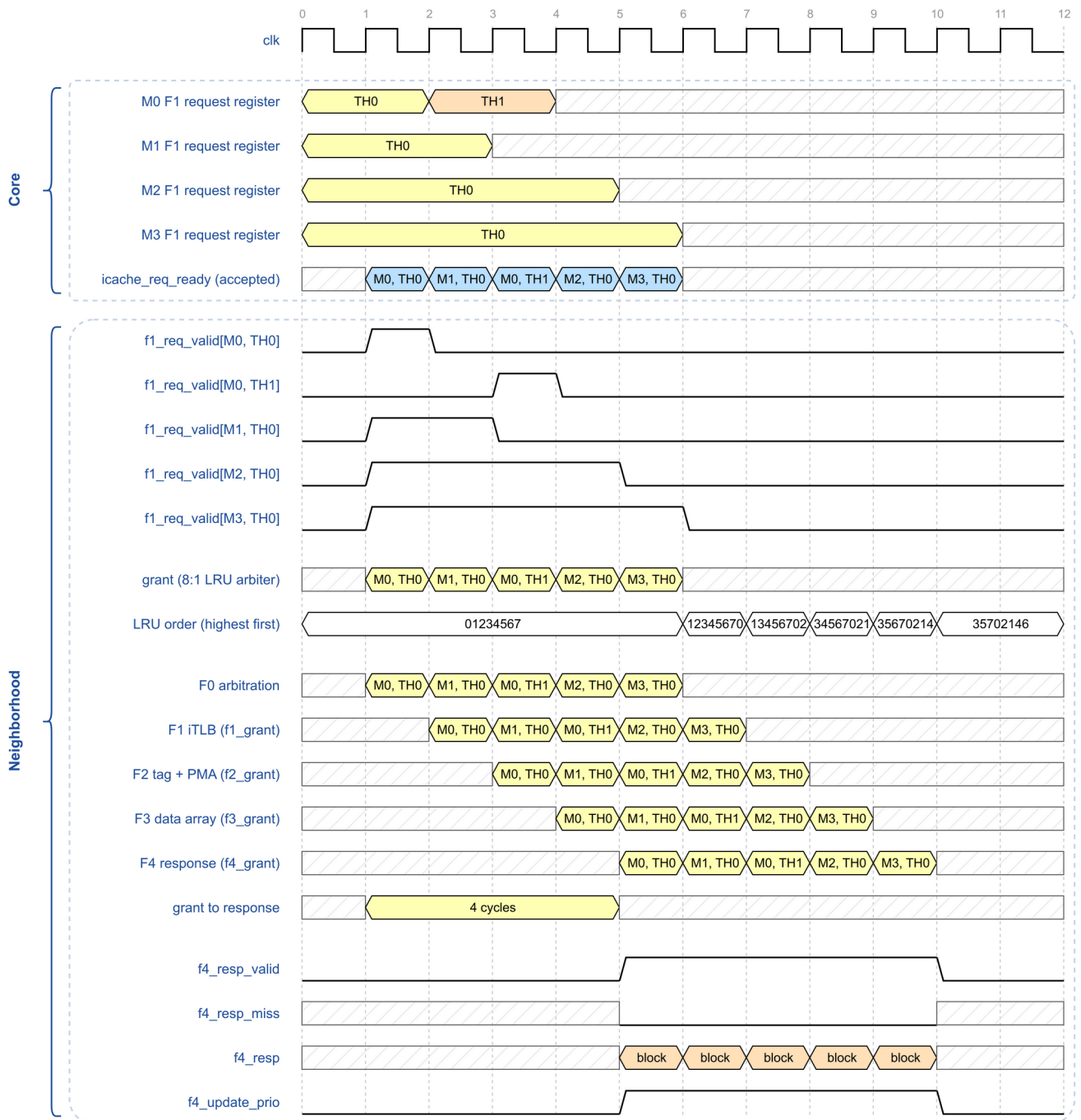


Figure 26. Request capture, arbitration and response for four cores

In cycle 0 the four cores present a TH0 request each, and the F1 registers hold them. The thread registers go valid in cycle 1, M0, TH0 wins and **icache_req_ready** frees M0's F1 register, which reloads with TH1 in cycle 2. M1, TH0 is granted in cycle 2, and in cycle 3 M0, TH1 is granted, while M2 and M3 are still holding their first request.

M0 is therefore served twice before M2 and M3 are served once, and the reason is that the aging matrix has one entry per thread, not one per Minion. Granting M0, TH0 demotes M0, TH0 and nothing else, so M0, TH1 keeps a position that is still ahead of M2 and M3 and wins the arbitration in cycle 3. The deferred update does not cause this: had the priority been updated at the grant instead of at the response, M0, TH1 would have won that cycle just the same.

Per-thread entries are what make the arbitration fair, because the two threads of a core succeed and fail independently. One may hit while the other misses, and a single entry shared by both

would demote the thread that missed every time its sibling was served, costing it the position it needs to win its retry after the fill. What the deferred update adds is the next step: a thread whose own response is a miss keeps its place, so it wins its retry immediately, and the miss-count guard bounds how long any thread can be passed over (Section 3.3.8).

The response for M0, TH0 appears in F4 in cycle 5, four cycles after its grant, with `f4_grant` naming the thread. It hits, `f4_update_prio` is asserted and the order changes from `01234567` to `12345670`.

3.3.8. Livelock and Starvation Avoidance

Up to eight threads share each L0 micro-cache, and the two threads of a core share its fetch port, decoders and issue slot. At every shared point one thread can be delayed by the others. Each shared point therefore has a policy that limits how long a thread can be held back: LRU or round-robin arbitration, a priority rule that favours threads whose requests missed, a wake-up protocol that never loses a wake-up, and a miss-count guard that gives a starving thread exclusive access to the L0 micro-cache. Table 30 lists these mechanisms, and the subsections that follow describe them.

Figure 27 adds to the interface of Figure 23 the two blocks that carry the guard: the per-thread miss counters, one set per micro-cache, and the mask they drive, which gates the bids into the 8:1 LRU arbiter.

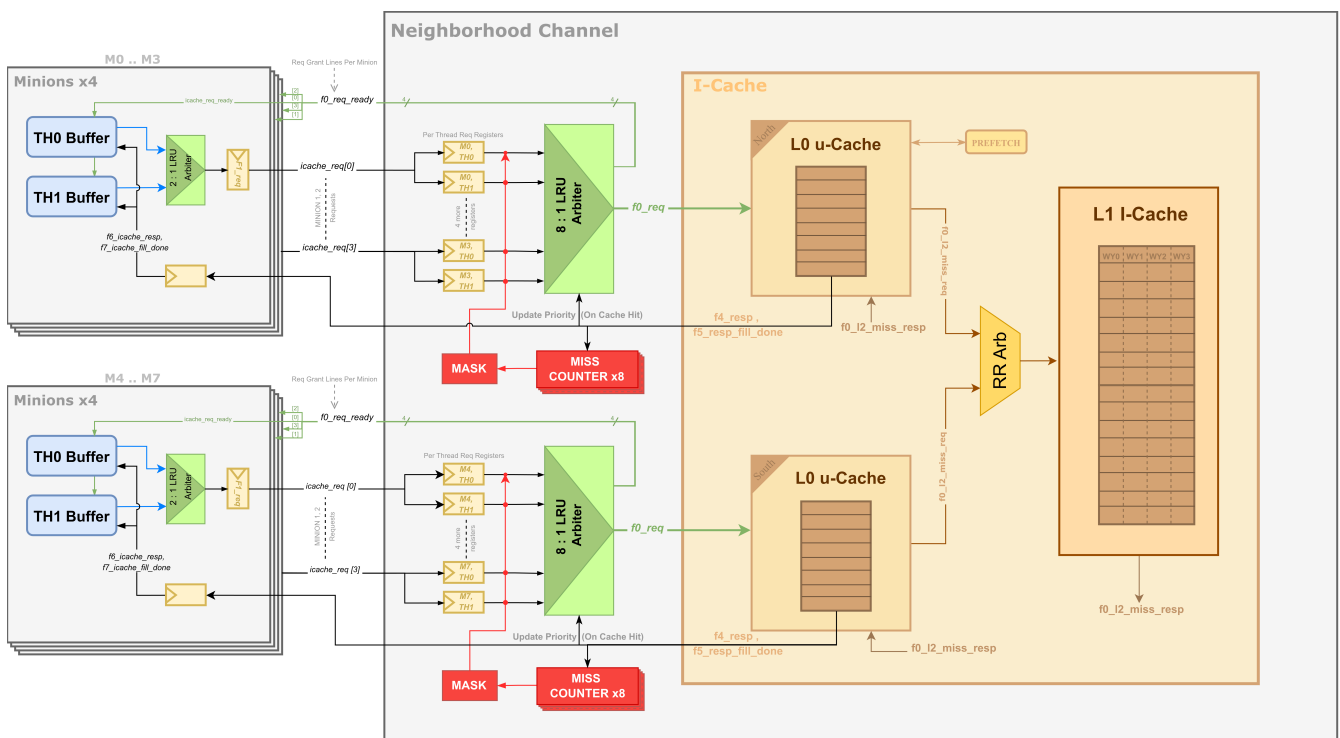


Figure 27. Miss counters and bid mask in the fetch path

Table 30. Starvation avoidance mechanisms in the instruction fetch path

Shared resource	Contenders	Mechanism	Guarantee
Core fetch port (F1 request register)	2 threads of a core	2:1 LRU arbiter, priority updated on every grant	A waiting thread is granted after at most one request of the other thread

Shared resource	Contenders	Mechanism	Guarantee
L0 request arbiter	8 threads of 4 cores	8:1 LRU arbiter whose priority is updated only on a hit or a fault	A thread whose request missed keeps its priority for its retry
L0 miss handler	8 threads and the prefetcher	Fill-done broadcast with ordered timing; thread wakes on fill done, redirect, disable or halt	A thread sleeps only while a fill is in progress, and always wakes
L0 miss handler (repeated misses)	8 threads	Per-thread consecutive-miss counters and bid mask	A thread with 31 consecutive misses gets exclusive access for its core until it hits
L0 miss handler (prefetch)	Prefetcher and threads	Prefetch requests enter the L0 micro-cache only in cycles with no core request	Demand fetches always win the L0 pipeline over prefetch
L0 miss handler (retry windows)	Prefetcher and waking threads	8-cycle prefetch pause after each fill done; prefetch blocked after an unattended miss until the next fill done, then until a core request or 16 cycles	Waking threads get the first chance at the freed miss handler, and prefetching is never blocked for good
L1 I-cache miss port	2 L0 micro-caches	Round-robin arbiter	Each L0 micro-cache waits for at most one miss of the other
Decoders and ID issue slot	2 threads of a core	Alternating decode selection and round-robin thread scheduler, per-thread FIFOs and stalls	Each eligible thread issues at least every other cycle

The mechanisms in the first two rows are described with the hardware that implements them: the 2:1 LRU arbiter of the core fetch port in [Section 2.2.3](#), and the 8:1 LRU arbiter with its deferred priority update in [Section 3.3.3](#). The code prefetcher and the two timed windows that hold it behind every thread are described in [Section 3.4.10](#), the round-robin arbiter between the two micro-caches in [Section 3.5.5](#), and the decoders, instruction FIFOs and thread scheduler in [Section 2.2.7](#) and [Section 2.2.9](#). What remains, and what the rest of this section describes, is the one shared resource with no arbitration of its own: the miss handler of the L0 micro-cache.

3.3.8.1. The L0 Miss Handler

The miss handler is described in full in [Section 3.4.8](#). Enough of it is repeated here for this section to stand on its own.

Each L0 micro-cache has one miss handler: the logic that handles the L0 micro-cache's single outstanding request to the L1 instruction cache. It consists of the following:

1. a three-state machine (`f0_miss_state`: Ready, Request, Fill_Wait),

2. a register holding the physical address of the missing line (`f0_miss_addr`), and
3. a flag indicating the miss as a prefetch miss (`f0_miss_is_prefetch`).

A miss is accepted only in the Ready state. The handler then sends the request to the L1 instruction cache (Request state), waits for the line (Fill_Wait state), writes it into the L0 micro-cache unless it is a prefetch, and returns to the Ready state, at which point fill done is generated.

Behind it, the L1 instruction cache is also single-transaction. It accepts one request, closes its request port until that request's response, and always answers with the block: on an L1 hit from its data RAM, on an L1 miss once the line has come back from the Shire Cache. A micro-cache miss that reaches the L1 therefore occupies the L1 for its whole duration, which on an L1 miss is a full round trip out of the Neighborhood.

The code prefetcher shares this same handler. It never delays a fetch in the same cycle, because a prefetch request enters the L0 pipeline only in cycles with no core request, but once it has taken the handler it holds it exactly as a demand miss would ([Section 3.4.10](#)).

A lookup that does not hit returns a miss response to the thread in one of the following three cases:

1. The line is not present in the L0 micro-cache and the handler is idle, so the handler accepts the miss and sends it to the L1 instruction cache.
2. The line is not present in the L0 micro-cache but the handler is busy with another miss, so the miss is not recorded and no request is sent to the L1 instruction cache for it.
3. The address misses in the L0 TLB, so a translation is requested from the page-table walker ([Section 3.6](#)) and no request is sent to the L1 instruction cache for it.

In all three cases the thread enters the sleep state (`f0_miss_pending`) and no request is issued. In the second and third nothing has been requested on that thread's behalf, and it is the second that this section is about.

3.3.8.2. Sleep and Wake-Up

The L0 micro-cache pulses `f6_icache_fill_done` to all its threads whenever an L1 fill completes, whether it is a demand miss or a prefetch, and whenever a translation fill completes. On seeing the fill done every sleeping thread wakes and retries according to the arbitration priority. A thread whose line is now present hits, including any thread that missed on the line already being fetched, and a thread whose line is still absent misses again; if the handler is now idle, its miss is sent to the L1 instruction cache.

A thread sleeps because its own response was a miss; that is the only thing that puts it to sleep. What decides when it is woken is what the handler was doing at that moment: serving this thread's own miss, serving the fill of another thread that kept the handler busy, or completing a translation. In all three a fill is in progress, so every sleeping thread is woken when it receives the fill-done signal. The fill-done timing ([Section 2.2.4.3](#)) makes sure that no wake-up is lost: fill done is issued one cycle after the last response the fill can affect, and it is registered again in the Front End. A redirect, a thread disable or a pending halt clears the sleep state of the affected thread only; fill done is the only event that wakes all sleeping threads.

Which thread is served in a round is decided in F2. That is the stage where the tag is compared,

where a miss is therefore first indicated, and where the handler accepts it and flops its state so that the request to the L1 instruction cache goes out in F3 (Table 29). A lookup that reaches F2 any later in the round finds the handler no longer Ready.

Two properties of this protocol follow, and they set up the rest of the section. Because the handler accepts a miss only while it is Ready, and because it is Ready only between one fill completing and the next miss being accepted, each fill-done round serves exactly one new miss. And because the whole set of sleeping threads is woken at once, the thread that is served in a round is whichever one reaches F2 first with a miss.

3.3.8.3. How a Thread Can Be Left Behind

The thread that reaches F2 first is not always the thread that waited longest. A thread that was woken by something other than fill done is already moving while the others are still asleep, and it keeps that head start all the way down the pipeline.

The following sequence is built from ordinary events and needs no fault, no error and no unusual configuration.

Starting point. Each fetch returns a 32-byte block, half an L0 line, and a thread buffer holds two of them (Section 2.2.5). Thread A has the first half of line X in its buffer and is executing from it. It requests the second half of X. While A was executing, another thread's fill selected X as its LRU victim and evicted it, so A's request misses. The miss handler is busy with that other fill, so A's miss is **not recorded**. A receives a miss response and sleeps.

The redirect. While A sleeps, the integer pipeline resolves a branch in the instructions A already consumed and redirects it. The redirect clears A's sleep bit at once, and A presents a request for the branch target without waiting for any fill.

The capture. A's new request is in F1 when `icache_fill_done` arrives for the other thread's line. The seven other threads wake on that pulse, but they are a cycle behind: they must be granted before they enter F0, while A is already one stage inside the pipeline. A reaches F2 first, misses on the branch target, finds the handler just returned to **Ready**, and takes it.

The others. Each of the seven woken threads reaches F2 one or more cycles later and finds the handler in **Request** or **Fill_Wait**. Every one of their misses is discarded. They receive miss responses, sleep again, and because a miss response does not update the arbitration priority, the next round begins in exactly the same order as this one.

Thread A has not done anything wrong, and no arbiter has misbehaved: A was granted on merit and its miss was accepted on merit. The point is that a redirect let A bypass the queue of sleeping threads entirely, and that one fill-done round can serve only one miss. A thread that is redirected often enough, or a prefetch engine that is already in the pipeline when the threads wake, can take that single opportunity repeatedly.

The result is a thread that is granted in every round, answered in every round, and never once has a line fetched on its behalf. Its arbitration priority does not change, because priority is only updated on a hit or a fault, so the arbiter cannot lift it out of the position it is in. Raising its priority would not be enough in any case: priority decides who enters the pipeline first, and the thread that enters first is not necessarily the thread that finds the handler idle.

This is the case the miss-count guard exists for.

3.3.8.4. Miss-Count Livelock Guard

The Neighborhood counts, for each thread, the number of consecutive miss responses to that thread's own requests (`th_miss_cnt`, 5 bits, saturating at 31). The counter advances only when that thread's own response returns, so a thread that is not being granted does not accumulate.

While no counter is saturated, every thread may bid. When one or more saturate, a mask register (`th_miss_mask`) allows only the two threads of one Minion: the lowest-numbered Minion that has a saturated thread. Both of its threads are allowed, because the Minion holds one request at a time and the sibling's request may have to be accepted before the starving thread's can be presented at all. The mask is recomputed one cycle after every response.

The mask does not discard anything. A bid is the thread's request-register valid qualified with the mask, so masking only removes the thread's valid at the arbiter: the request itself stays in its register, untouched, and bids again in the cycle the mask is released. Nothing is re-issued and no state is lost.

With every other Minion masked, only the selected Minion's misses can reach the handler, so the starving thread's line is fetched and cannot be displaced before the thread retries. The thread then hits, which clears its counter, and the mask passes to the next Minion with a saturated thread or is removed if there is none. The mask is released by a hit, and only by a hit.

Figure 28 shows the guard at work, with the waits between rounds compressed into breaks. Every miss costs M0, TH0 a full round: the miss response puts it to sleep, fill done wakes it, it requests again and is granted, and its next response arrives four cycles after that grant.

The first response shown takes the counter to 31. The counter is loaded at the end of that cycle, so `th_miss_cnt_sat` only reads one in the next, and `th_miss_mask_en` is a registered copy of the counter enable, so the mask itself changes one cycle after that: the restrictive mask is in force two cycles after the response that saturated the counter. From that point only the two threads of Minion 0 bid.

Being alone does not give M0, TH0 its line, it gives it the miss handler. On the next round its lookup is the only one to reach F2, so its miss is accepted: `f0_l1_miss_req_valid` is driven in F3 and `f0_l1_miss_req` carries that thread's line address out to the L1 instruction cache. The response in F4 is still a miss and the thread sleeps again; the counter holds at 31 because it saturates. Only when that fill completes and the thread retries does it hit, and that hit clears the counter and releases the mask.

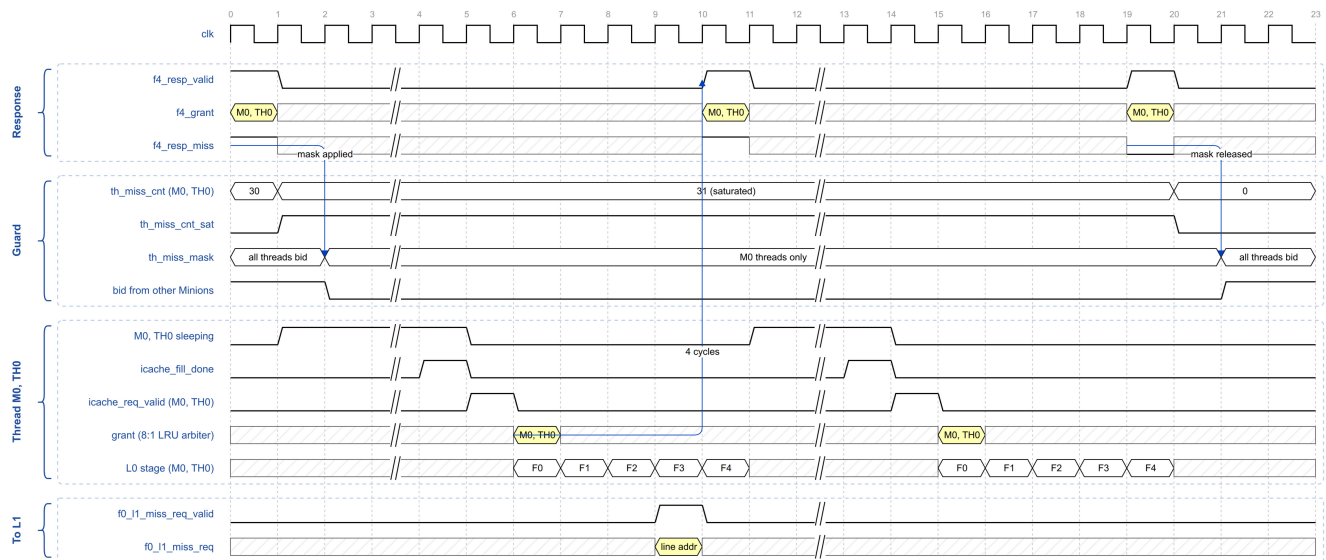


Figure 28. Miss-count guard: saturation, bid mask and release

Counters and Error Responses

A response updates the counters of more threads than the one it belongs to. [Table 31](#) lists the cases.

Table 31. Effect of a response on the per-thread miss counters

Response returned in F4	Counter of the thread it belongs to	Counters of the other seven threads
Hit	Cleared	Unchanged
Miss, no fault or error	Incremented, saturating at 31	Unchanged
Page fault or access fault	Cleared	Cleared
Bus error or ECC error	Cleared	Cleared

The first two rows are the counter doing its job. The last two are deliberate: an error clears every thread's counter, not only the counter of the thread that received it.

The reason is a corner case in which a core would otherwise hang on a faulting address. A faulting thread never hits on that address, so its counter would saturate and it would hold exclusive access, which is only ever released by a hit. Clearing the counters and the mask lets the fault be delivered and handled normally rather than leaving a mask in place based on stale counts. If the fetch pattern that caused the pressure persists, the counters saturate again and the guard is triggered again.

The cost of clearing all eight is stated in [Section 3.3.8.5](#): a sustained stream of faulting fetches from any thread defers the guard for all of them.

3.3.8.5. Assumptions

These mechanisms guarantee forward progress in the fetch path under the following conditions:

- Every L1 miss request eventually returns a fill, and every translation request eventually completes.

- Software does not keep a thread's exclusive mode, **WFI** or other architectural stalls asserted indefinitely. These deliberately stop the affected thread and are not starvation.
- Error responses are not sustained. Because an error clears every thread's miss counter, a continuous stream of faulting fetches from other threads would defer the livelock guard. In practice each fault redirects the faulting thread to its trap handler, so such a stream does not persist.

3.3.9. Known Limitations

The properties below follow from the design of the interface. They are not defects, but they constrain how the interface may be changed and how it behaves under load.

- **Response carries no thread identifier.** A core matches a response to its request by counting cycles, so the fixed response latency is part of the interface contract. A change to the depth of the L0 pipeline has to be matched in the Front End.
- **Fill done carries no address.** Every fill wakes every sleeping thread of the micro-cache. Threads waiting for a different line arbitrate, miss again and sleep again, which costs L0 bandwidth on every fill and grows with the number of threads waiting on different lines.
- **One outstanding miss per L0 micro-cache.** A miss that arrives while the miss handler is busy is answered with a miss response but is not recorded, so its line is not requested. That thread makes progress only after an unrelated fill wakes it and it retries.
- **Starving Minions are served in a fixed order.** When several Minions have a saturated thread, the mask always selects the lowest-numbered one first. Its waiting time is bounded, because each exclusive period ends when the selected thread hits, but the order is not fair.
- **The L1 instruction cache is blocking.** It accepts one request at a time and holds its request port closed until that request's own response, which on an L1 miss waits for the line to come back from the Shire Cache. The round-robin bound of one transaction of the other micro-cache is therefore correct as a count but stands for a full round trip out of the Neighborhood, during which the other micro-cache cannot even begin a lookup that might have hit.

3.4. L0 Micro-Cache

3.4.1. Overview

Table: L0 micro-cache characteristics.

3.4.2. Internal Blocks

Figure: L0 micro-cache block diagram.

3.4.3. Pipeline Stages

Figure: L0 micro-cache pipeline.

3.4.4. Instruction TLB

3.4.5. Micro-Tag and Micro-Data Arrays

3.4.6. PMA Check

3.4.7. Replacement

3.4.8. Miss Handling

3.4.8.1. TLB Miss

3.4.8.2. Tag Miss

3.4.8.3. Fill and Thread Wake-Up

3.4.9. L1 Bypass

3.4.10. Software Prefetch

Figure: Prefetch state machine.

3.5. L1 Instruction Cache

3.5.1. Overview

Figure: L1 instruction cache block diagram.

3.5.2. Organization

3.5.3. Tag, LRU and Data Arrays

3.5.4. External Data RAM

3.5.5. Request Arbitration Between L0 Micro-Caches

3.5.6. Miss Path

3.5.7. Errors and ECC

3.6. Shared Page-Table Walker

3.6.1. Clients and Arbitration

3.6.2. Walk Sequence and Page Sizes

Figure: Page-table walk state machine.

3.6.3. PTE Cache and Invalidation

3.7. Memory Request and Response Path

3.7.1. Miss, Evict and Fill FIFOs

Figure: Neighborhood memory request and response path.

3.7.2. Request Arbitration

3.7.3. Response Routing and Backpressure

3.8. Tensor Support

3.8.1. Cooperative Tensor Load

3.8.2. Tensor Store Buffer

3.8.3. TBOX Router

3.9. Synchronization, Messaging, Registers and PMU

3.9.1. Fast Local Barrier

3.9.2. Local Message Network

3.9.3. IPI and FCC Paths

3.9.4. Neighborhood Registers and APB Access

3.9.5. Debug Channel

3.9.6. Performance Monitoring Unit